Supporting Information: Unraveling Chronic Disease Risk Evolution through Major Two-Feature Interaction Effects across Heterogeneous Subpopulations in the U.S.

1 Results

1.1 The Evolution of Age Group 1: Negative Risk Patterns

Focusing on the stratum where GenHL = 1 within Age Group = 1, we identify four major 1-feature categories associated with negative disease risk. These features are highlighted by bold-faced, boxed identifiers along the left vertical axis of the first column panel in Figure 1. Among these, $Veggies_1$ is particularly notable: a blue arrow connects it to $HighBP_0$ on the right vertical axis, indicating a confirmed interaction effect between $(Veggies_1, HighBP_0)$. This interaction is symmetric, as evidenced by a reciprocal blue arrow originating from $HighBP_0$ (also a major 1-feature category with negative disease risk) pointing back to $Veggies_1$. A similar symmetric pattern is also observed between $HighBP_0$ and $HighChol_0$.

However, such symmetry is not universally observed among the order-2 interaction effects in this panel. For example, $EDUgp_2$—another major 1-feature category associated with negative disease risk—contributes additional information when combined with $HighBP_0$ and $HighChol_0$. Yet, the reverse is not true: neither $HighBP_0$ nor $HighChol_0$ provide incremental information when conditioned on $EDUgp_2$. This asymmetry highlights the inherently directional nature of certain order-2 interaction effects.

An even more intriguing case arises with $Smoker_1$. Although it is not classified as a major 1-feature category and is typically associated with positive disease risk, it contributes negatively when interacting with $HighChol_0$. This directional order-2 effect—$(HighChol_0, Smoker_1)$—thus reflects a context-specific reversal of expected risk patterns.

Lastly, $PhysActivity_1$ represents a relatively typical pattern. Although not a major 1-feature category, it contributes additional information to $HighBP_0$, forming a directional interaction that aligns with its usual association with lower disease risk.

As we transition to the stratum where GenHL = 2 within Age Group = 1, the second column panel of Figure 1 reveals a distinct set of four major 1-feature categories associated with negative disease risk, differing notably from those observed under GenHL = 1. In this slightly less healthy subgroup, both $DiffWalk_0$ and $NoDocbcCost_0$ exhibit symmetric order-2 interaction effects indicative of negative disease risk. These two features also engage in directional order-2 interactions with $PhysActivity_0$—itself a major 1-feature category—where $PhysActivity_0$ contributes additional information to both $DiffWalk_0$ and $NoDocbcCost_0$, but not vice versa.

A particularly noteworthy feature of this panel is the presence of three "green arrows," which indicate transitions from negative to positive disease risk. Specifically, when $DiffWalk_0$, a negative-risk feature, is paired with $Incomegp_1$ (representing the lowest income category) or with $NoDocbcCost_1$, the resulting order-2 interactions—$(DiffWalk_0, Incomegp_1)$ and $(DiffWalk_0, NoDocbcCost_1)$—switch to positive disease risk. A third such transition is observed in the interaction $(NoDocbcCost_0, DiffWalk_1)$.

Interestingly, although $HighChol_0$ is typically associated with positive disease risk, in this context it contributes additional information to $NoDocbcCost_0$ without causing a risk polarity reversal; the interaction remains within the negative risk domain. A similar pattern is observed with Sex_1, which also provides extra information to $NoDocbcCost_0$ and to another major negative-risk feature, $Incomegp_3$ (the highest income category), thereby forming directional order-2 interactions that preserve the negative disease risk classification.

Finally, within this young and relatively healthy subgroup, $Fruits_1$ enhances the negative disease risk contributions of $NoDocbcCost_0$ and $Incomegp_3$ through the directional interactions $(NoDocbcCost_0, Fruits_1)$ and $(Incomegp_3, Fruits_1)$, respectively. However, no such enhancing effect is observed in relation to $DiffWalk_0$ or $PhysActivity_0$.

As general health level (GenHL) deteriorates, the GenHL = 3 stratum within the youngest age group (Age = 1) gives rise to four major 1-feature categories associated with negative disease risk: $Smoker_0$, $DiffWalk_0$, $Diabetes_0$, and $PhyHlthgp_1$. These features are indicated by bold-faced boxed identifiers along the left vertical axis of the third column panel in Figure 1. Notably, no symmetric order-2 interaction effects are observed among these four categories. Instead, only directional effects are present, including $(DiffWalk_0, Smoker_0)$, $(DiffWalk_0, PhysActivity_1)$, $(Diabetes_0, PhysActivity_1)$, $(Diabetes_0, Smoker_0)$, and $(PhysActivity_1, Smoker_0)$. Particularly interesting is the role of $Smoker_0$, which frequently acts as an enhancing feature within these directional interactions, with $PhysActivity_1$ playing a more moderate supportive role. Additional reinforcement of negative risk is provided by $HighBP_0$ and $EDUgp_3$, which respectively enhance the effects of $Smoker_0$ and $DiffWalk_0$.

Advancing to GenHL = 4, the stratum just prior to the worst health category, the number of major 1-feature categories declines further, leaving only two: $Smoker_0$ and $HighBP_0$. Despite the reduced number, both features form symmetric order-2 interaction effects indicative of negative disease risk. Moreover, $Smoker_0$ is further enhanced by $CholCheck_0$, and $HighBP_0$ by $Incomegp_3$. However, an important shift occurs when $Smoker_0$ is paired with $HighBP_1$, transitioning from negative to positive disease risk. This polarity reversal signals a context-specific shift in interaction dynamics.

Finally, in the most compromised health stratum—GenHL = 5—within this youngest age group, the number of major 1-feature categories with negative disease risk increases to seven. Among them, $PhyHlthgp_1$ and $Incomegp_3$ do not participate in any order-2 interaction effects, either symmetric or directional. The remaining five features are well-established contributors to negative disease risk and collectively exhibit a complex network of interactions. Several symmetric order-2 effects are evident, such as $(HighBP_0, DiffWalk_0)$, $(HighChol_0, DiffWalk_0)$, and $(Smoker_0, DiffWalk_0)$.

Directional order-2 effects provide further insight into the disease risk dynamics. For instance, the interaction $(HighChol_0, HighBP_0)$ is confirmed, whereas the reverse $(HighBP_0, HighChol_0)$ is not. This asymmetry suggests that the effect of $HighBP_0$ is more dominant in influencing disease risk than $HighChol_0$. Extending this interpretation, we observe that the negative disease risk associated with $HighChol_0$ is further amplified in the interactions $(HighChol_0, Incomegp_3)$ and $(HighChol_0, CholCheck_0)$. Similarly, $Smoker_0$ exerts increased influence through its interaction with $HighBP_0$, as captured by $(Smoker_0, HighBP_0)$.

It is particularly noteworthy that the negative disease risk associated with $Diabetes_0$ can be reinforced through multiple directional order-2 interaction effects. Specifically, five such interactions are observed: $(Diabetes_0, DiffWalk_0)$, $(Diabetes_0, CholCheck_0)$, $(Diabetes_0, Incomegp_3)$, $(Diabetes_0, HighBP_0)$, and $(Diabetes_0, HighChol_0)$. These results indicate that $Diabetes_0$ acts as a central node within the network of risk-lowering interactions in this subgroup. However, the effect of $Diabetes_0$ is not uniformly stable—when paired with $HighChol_1$, a reversal occurs. The directional interaction $(Diabetes_0, HighChol_1)$ leads to a switch from negative to positive disease risk, highlighting a context-dependent vulnerability in its protective role.

In addition to the three previously identified symmetric order-2 effects, the negative disease risk of $DiffWalk_0$ is further enhanced through the directional interaction $(DiffWalk_0, PhyHlthgp_2)$. This suggests that deteriorating self-reported physical health can amplify the protective effect of mobility-related functionality.

1.2 The Evolution of Age Group 1: Positive Disease Risk Patterns

Within the Age = 1 subpopulation, the evolution of positive disease risk, captured across the five column panels of Figure 2, progresses sparsely from GenHL = 1 through GenHL = 4, before exhibiting a drastic phase transition at GenHL = 5. The key developments across these strata are summarized below.

At GenHL = 1, only three major 1-feature categories associated with positive disease risk are identified: $HighBP_1$, $HighChol_1$, and the less conventional $Veggies_0$. These are marked as bold-faced box IDs in the first column panel of Figure 2. Although $HighBP_1$ and $HighChol_1$ are established risk factors, $Veggies_0$ usually considered a healthy trait, plays an enhancer role here. This is evident in the directional interaction $(HighBP_1, Veggies_0)$, where $Veggies_0$ contributes additional risk information to $HighBP_1$, but not vice versa. Similarly, $Incomegp_1$ enhances $HighChol_1$ via the interaction $(HighChol_1, Incomegp_1)$. In contrast, reducer roles are observed in $Diabetes_0$ acting upon $HighChol_1$, and both $EDUgp_3$ and $PhysActivity_1$ acting upon $Veggies_0$. These directional interactions, such as $(Veggies_0, EDUgp_3)$ and $(Veggies_0, PhysActivity_1)$, do not reverse the risk classification but mitigate its intensity. For example, $Diabetes_0$ reduces, but does not, the positive disease risk imposed by $HighChol_1$.

At GenHL = 2, three less commonly recognized 1-feature categories emerge. A symmetric order-2 interaction is observed between $PhysActivity_1$ and $Incomegp_1$, suggesting mutual enhancement. In addition, $NoDocbcCost_1$ forms a directional enhancer relationship with $PhysActivity_1$ via the interaction $(PhysActivity_1, NoDocbcCost_1)$. In particular, a surprising reversal in disease risk is observed: the interaction $(PhysActivity_1, Incomegp_3)$ results in a positive-to-negative switch, marked by a green arrow. This indicates a contextual reduction in the risk of disease driven by a higher income level.

At GenHL = 3, four major 1-feature categories of positive disease risk are identified, including the less common $PhyHlthgp_2$. No symmetric interactions are found; only directional ones are present. For example, $(DiffWalk_1, EDUgp_3)$ reflects the mitigating effect of higher education. In contrast, $Diabetes_1$ has three directional enhancers: $HighBP_1$, $DiffWalk_1$, and $Smoker_1$. $Smoker_1$ itself is enhanced by $DiffWalk_1$ but reduced by $MentHlthgp_4$. Likewise, $PhyHlthgp_2$ receives enhancement from $Smoker_1$ and $Diabetes_1$, but is mitigated by $MentHlthgp_4$.

At GenHL = 4, the risk structure becomes sparse. Only one major 1-feature category, $HighBP_1$, is detected, accompanied by a single 2-feature interaction: $(HighBP_1, BMIgh_3)$, where $BMIgh_3$ acts as an enhancer. This sparsity suggests that remaining in GenHL = 4 may serve as a protective

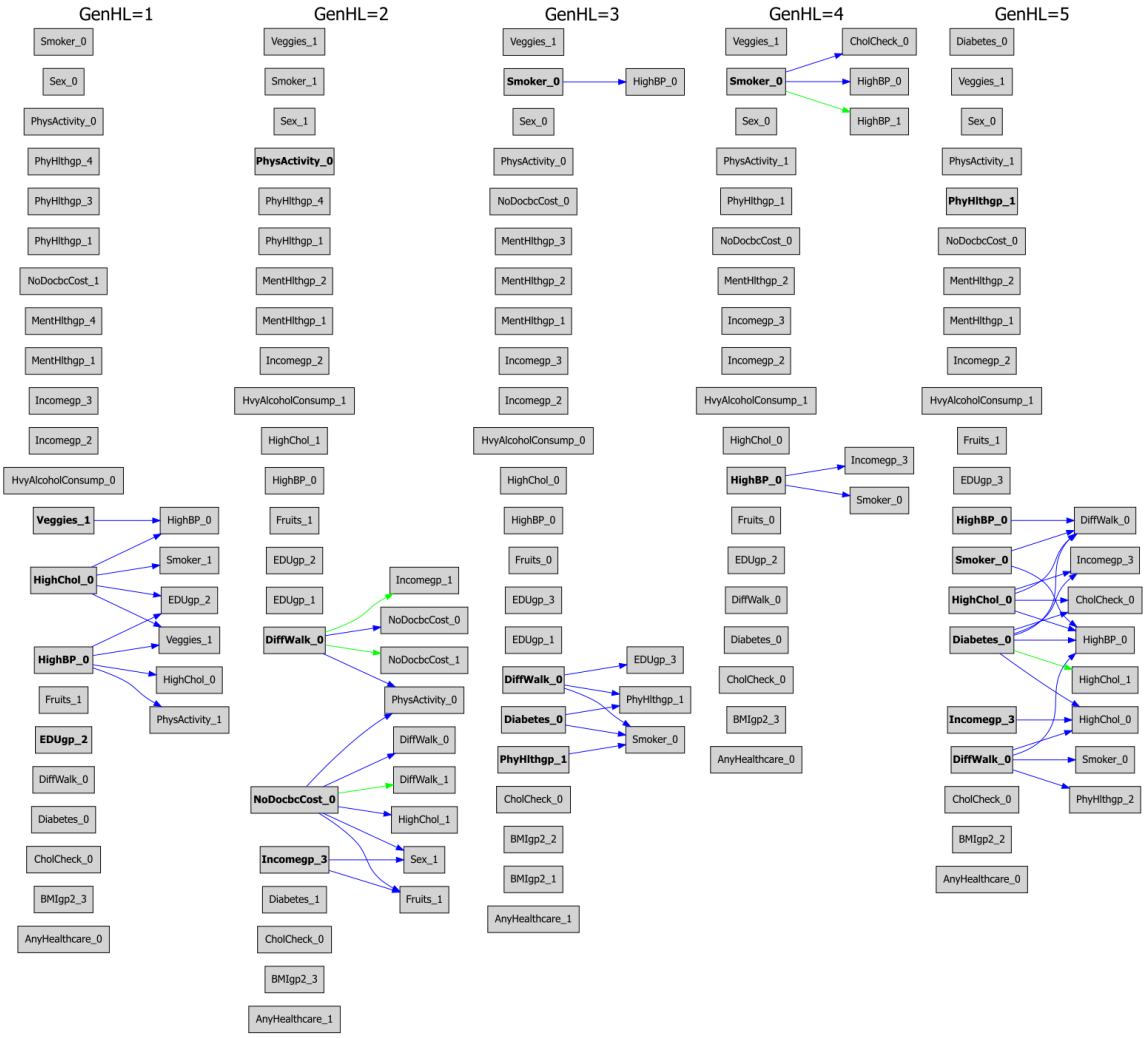

Figure 1: Age1's order-2 interacting effect evolution of negative chronic disease risk across GenHL: 1 to 5.

threshold; further deterioration to GenHL = 5 significantly increases chronic disease risk potential.

Indeed, at GenHL = 5, a sharp increase in both order-1 and order-2 effects is observed. Six major 1-feature categories are identified, all of which participate in symmetric order-2 interactions indicative of heightened disease risk. These include: $(Diabetes_1, Smoker_1)$, $(Diabetes_1, DiffWalk_1)$, $(HighBP_1, Diabetes_1)$, $(HighBP_1, HighChol_1)$, $(HighBP_1, Incomegp_1)$, $(HighBP_1, Smoker_1)$, $(HighChol_1, Smoker_1)$ and $(HighChol_1, DiffWalk_1)$. In addition, several directional enhancer interactions are observed: $(DiffWalk_1, HighBP_1)$, $(Incomegp_1, DiffWalk_1)$, and $(Incomegp_1, HighChol_1)$. Together, this dense network of mutual reinforcement signals a substantially elevated risk of chronic disease when multiple risk factors are present simultaneously.

However, it is also noteworthy that even within this highest-risk stratum, positive-to-negative switching is still possible. Both $Smoker_1$ and $Incomegp_1$ can be redirected toward reducing risk under certain interactions, such as: $(Smoker_1, HighBP_0)$, $(Smoker_1, HighChol_1)$, $(Smoker_1, PhyHlthgp_1)$, $(Incomegp_1, HighBP_0)$, and $(Incomegp_1, PhyHlthgp_1)$. These findings underscore a critical implication: even within the most adverse health status (GenHL = 5), strategic combinations of lifestyle or socioeconomic factors can still suppress the potential for chronic disease development.

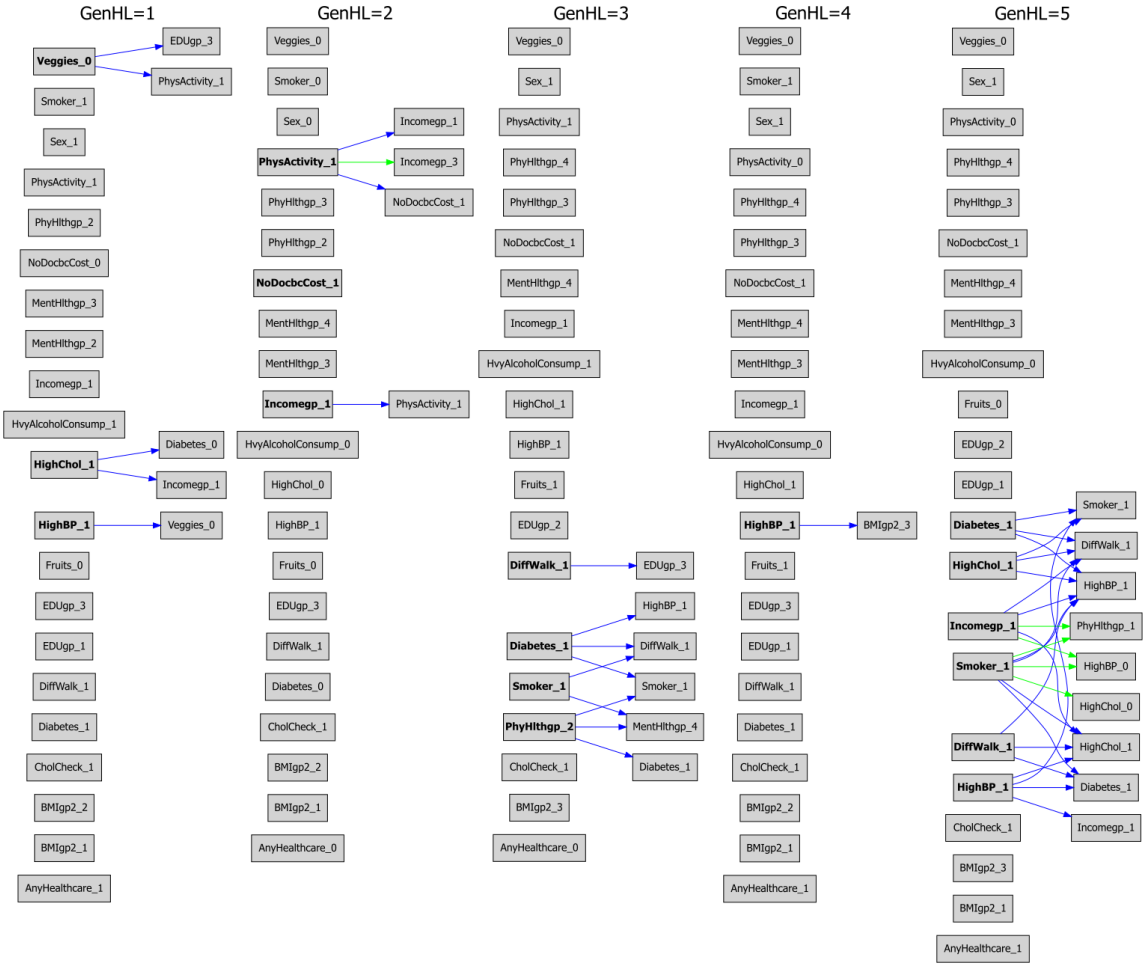

Figure 2: Age1's order-2 interacting effect evolution of positive chronic disease risk across GenHL: 1 to 5.

1.3 The Evolution of Age Group 2: Negative Disease Risk Patterns

In the Age = 2 subpopulation, the evolution of negative disease risk begins with a sparse presence of major 1-feature and 2-feature categories during the GenHL = 1 and GenHL = 2 strata. This

sparsity transitions into an abundant configuration at GenHL = 3, followed by a moderate presence at GenHL = 4. Notably, there is no GenHL = 5 stratum observed in this age group. The detailed characteristics of this progression, as visualized in Figure 3, are summarized below.

At GenHL = 1, only three relatively well-known major 1-feature categories are identified: $HighBP_0$, $HighChol_0$, and $DiffWalk_0$. Among them, $HighBP_0$ and $DiffWalk_0$ form a symmetric order-2 interaction indicative of negative disease risk. Additionally, two directional interactions are present: $(HighChol_0, HighBP_0)$, where $HighBP_0$ enhances $HighChol_0$, and $(DiffWalk_0, Smoker_0)$, with $Smoker_0$ contributing further reduction in disease risk. However, $DiffWalk_0$ can also be reversed to a positive disease risk through interactions with $HighBP_1$ and $Smoker_1$, namely $(DiffWalk_0, HighBP_1)$ and $(DiffWalk_0, Smoker_1)$.

At GenHL = 2, only two major 1-feature categories are found: $HighBP_0$ and $Incomegp_3$. Interestingly, these form a symmetric interaction $(HighBP_0, Incomegp_3)$, suggesting a mutual enhancement in reducing disease risk. This simplicity underscores the importance of these features in early-stage risk mitigation.

A significant transition occurs at GenHL = 3, where the number of major 1-feature categories increases to eight. These include $HighBP_0$, $HighChol_0$, $DiffWalk_0$, $PhyHlthgp_1$, $Diabetes_0$, $NoDocbcCost_0$, $Smoker_0$, and $Incomegp_3$. Among them, eight symmetric and fourteen directional order-2 interactions are detected. The symmetric effects include: $(HighBP_0, DiffWalk_0)$, $(HighBP_0, PhyHlthgp_1)$, $(HighChol_0, DiffWalk_0)$, $(HighChol_0, PhyHlthgp_1)$, $(Diabetes_0, DiffWalk_0)$, $(Diabetes_0, PhyHlthgp_1)$, $(Diabetes_0, NoDocbcCost_0)$ and $(Smoker_0, DiffWalk_0)$.

$Incomegp_3$ does not participate in any symmetric interaction but contributes directionally through $(Incomegp_3, HighBP_0)$. The remaining directional interactions include: $(HighChol_0, HighBP_0)$, $(HighChol_0, Incomegp_3)$, $(Smoker_0, HighBP_0)$, $(Smoker_0, PhyHlthgp_1)$, $(Smoker_0, HighChol_0)$, $(Diabetes_0, HighBP_0)$, $(Diabetes_0, Incomegp_3)$, $(Diabetes_0, MentHlthgp_1)$, $(Diabetes_0, HighChol_0)$, $(Diabetes_0, Smoker_0)$, $(NoDocbcCost_0, Smoker_0)$, $(DiffWalk_0, PhyHlthgp_1)$, $(DiffWalk_0, Incomegp_3)$ and $(DiffWalk_0, NoDocbcCost_0)$.

It is particularly noteworthy that $Diabetes_0$ is involved in three symmetric and five directional order-2 interactions, meaning it is influenced by eight other 1-feature categories. Yet, it is also overturned by six positive-risk features (discussed below), indicating its idiosyncratic role. This dual behavior suggests *Diabetes* may be more appropriately conceptualized as a component of chronic disease rather than a standalone marker. To a lesser extent, $DiffWalk$ exhibits a similar pattern—participating in four symmetric and two directional negative interactions, yet also experiencing five negative-to-positive risk reversals.

In total, six major 1-feature categories are subject to negative-to-positive switching through interactions with positive-risk features. Only $Incomegp_3$ and $PhyHlthgp_1$ remain stable. Examples include:

- $HighBP_0$ and $HighChol_0$ being overturned by $DiffWalk_1$;
- $Smoker_0$ by $HighBP_1$ and $MentHlthgp_4$;
- $NoDocbcCost_0$ by $DiffWalk_1$, $Incomegp_1$, and $HighBP_1$;
- $Diabetes_0$ by $DiffWalk_1$, $Incomegp_1$, $HighBP_1$, $PhyHlthgp_4$, $MentHlthgp_4$, and $HighChol_1$;
- $DiffWalk_0$ by $HighBP_1$, $PhyHlthgp_3$, $MentHlthgp_4$, $HighChol_1$, and $Diabetes_1$.

Each of these interactions forms a directional order-2 effect contributing to positive disease risk.

At GenHL = 4, the number of major 1-feature categories shrinks to six. Among these, only two symmetric interactions are found: $(HighBP_0, DiffWalk_0)$ and $(HighChol_0, DiffWalk_0)$. However, six directional effects remain: $(PhyHlthgp_1, HighBP_0)$, $(PhyHlthgp_1, DiffWalk_0)$, $(HighChol_0, HighBP_0)$, $(HighChol_0, Incomegp_2)$, $(HighChol_0, NoDocbcCost_0)$ and $(NoDocbcCost_0, PhyHlthgp_1)$.

Three negative-to-positive switches are also present in this stratum:

- $HighChol_0$ is overturned by $DiffWalk_1$;
- $NoDocbcCost_0$ is overturned by both $DiffWalk_1$ and $HighBP_1$.

Finally, it is important to emphasize that no GenHL = 5 stratum is observed in the Age = 2 subpopulation. This absence may reflect a natural health boundary within this age group, or it may suggest early-stage prevention or sampling limitations.

1.4 The Evolution of Age Group 2: Positive Disease Risk Patterns

In sharp contrast to the complex and dynamic structure of negative disease risk across the GenHL-axis, the evolution of positive disease risk in the Age = 2 subpopulation is markedly simpler, as illustrated in Figure 4. One particularly striking characteristic is the complete absence of positive-to-negative risk switching, a phenomenon observed in other age groups and in the negative risk domain, but entirely missing here.

At GenHL = 1, three major 1-feature categories are identified: $HighBP_1$, $HighChol_1$, and $Smoker_1$. Among these, two symmetric order-2 interactions are observed: $(HighBP_1, Smoker_1)$ and $(HighChol_1, Smoker_1)$. In addition, two directional interactions involving $HighChol_1$ are present: $(HighChol_1, DiffWalk_1)$ and $(HighChol_1, Incomegp_1)$.

In the GenHL = 2 stratum, the order-2 interactions of positive disease risk remain sparse. Among the three major 1-feature categories identified, only three 2-feature interactions are observed. One is symmetric—$(HighBP_1, Incomegp_1)$—while the other two are directional, including $(Smoker_1, HighBP_1)$. Notably, $Incomegp_1$ continues to appear as a major risk factor and is present across all strata along the GenHL-axis within this age group.

At GenHL = 3, the number of major 1-feature categories increases to five, again including $Incomegp_1$. The network of interactions becomes more complex, with six symmetric order-2 interactions detected: $(HighBP_1, HighChol_1)$, $(HighBP_1, Incomegp_1)$, $(HighBP_1, NoDocbcCost_1)$, $(HighChol_1, Incomegp_1)$, $(HighChol_1, NoDocbcCost_1)$ and $(Smoker_1, Incomegp_1)$.

In addition, ten directional order-2 interactions emerge: $(HighBP_1, NoDocbcCost_0)$, $(HighBP_1, DiffWalk_1)$, $(HighBP_1, MentHlthgp_4)$, $(HighChol_1, DiffWalk_1)$, $(HighChol_1, MentHlthgp_4)$, $(HighChol_1, PhyHlthgp_3)$, $(Smoker_1, HighBP_1)$, $(Smoker_1, HighChol_1)$, $(Smoker_1, DiffWalk_1)$ and $(Incomegp_1, DiffWalk_1)$.

At GenHL = 4, representing the terminal stage of the GenHL-axis within Age = 2, the structure of interactions simplifies once again. Among the five computed major 1-feature categories, only ten order-2 interactions are found. Six of these arise from three symmetric interactions: $(HighBP_1, Incomegp_1)$, $(HighBP_1, DiffWalk_1)$ and $(DiffWalk_1, Incomegp_1)$. The remaining four directional interactions include: $(HighBP_1, HighChol_1)$, $(HighChol_1, DiffWalk_1)$, $(NoDocbcCost_1, HighBP_1)$ and $(NoDocbcCost_1, DiffWalk_1)$.

An important observation is that $Incomegp_1$ functions in a role comparable to $HighBP_1$ and $HighChol_1$ throughout the GenHL evolution in this age group. It consistently appears as a major 1-feature category and frequently forms both symmetric and directional interactions. This consistent presence suggests that lower income may play an amplifying role in the aggregation of chronic disease risk among individuals in the Age = 2 group.

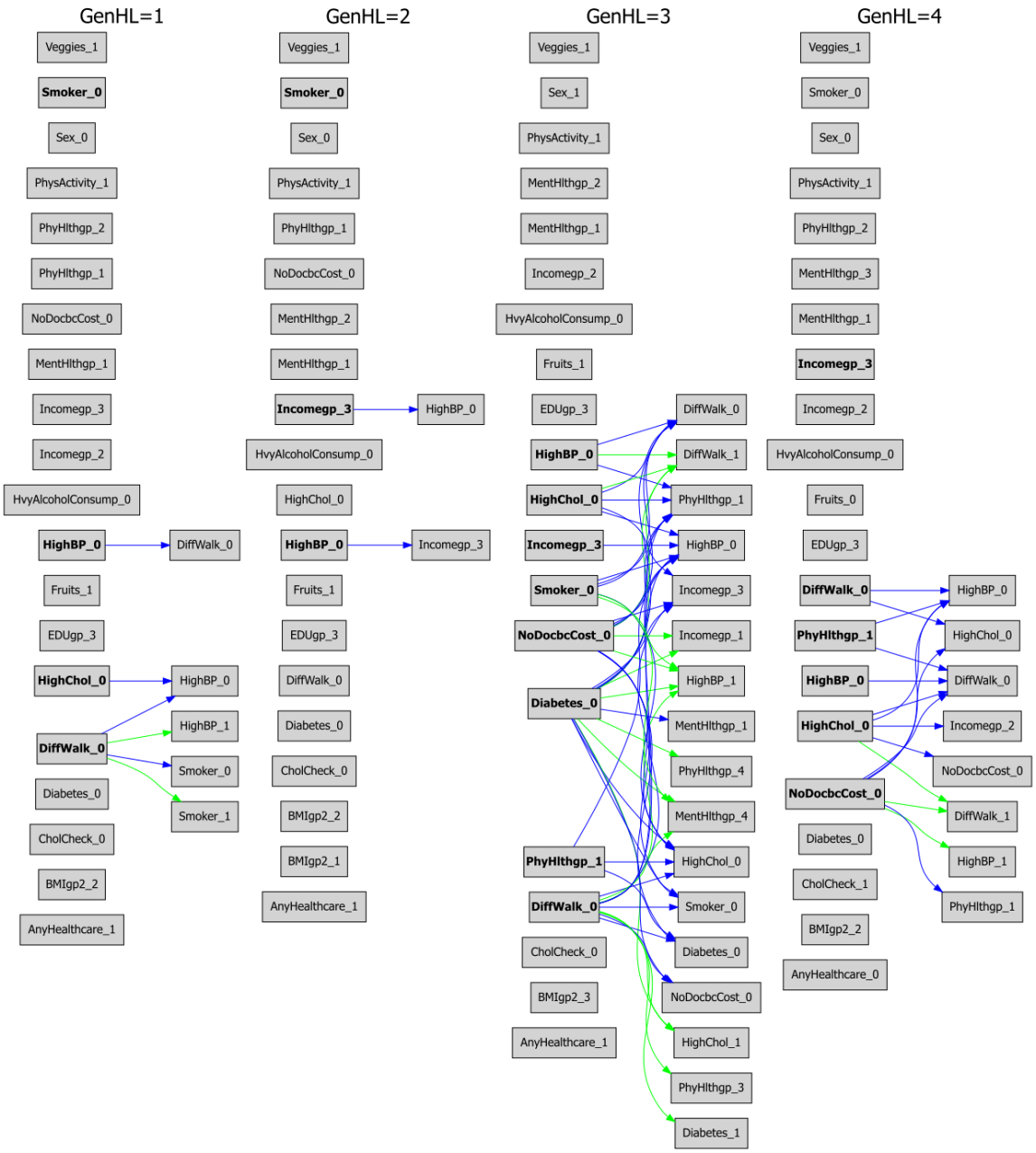

Figure 3: Age2's order-2 interacting effect evolution of negative chronic disease risk across GenHL: 1 to 4.

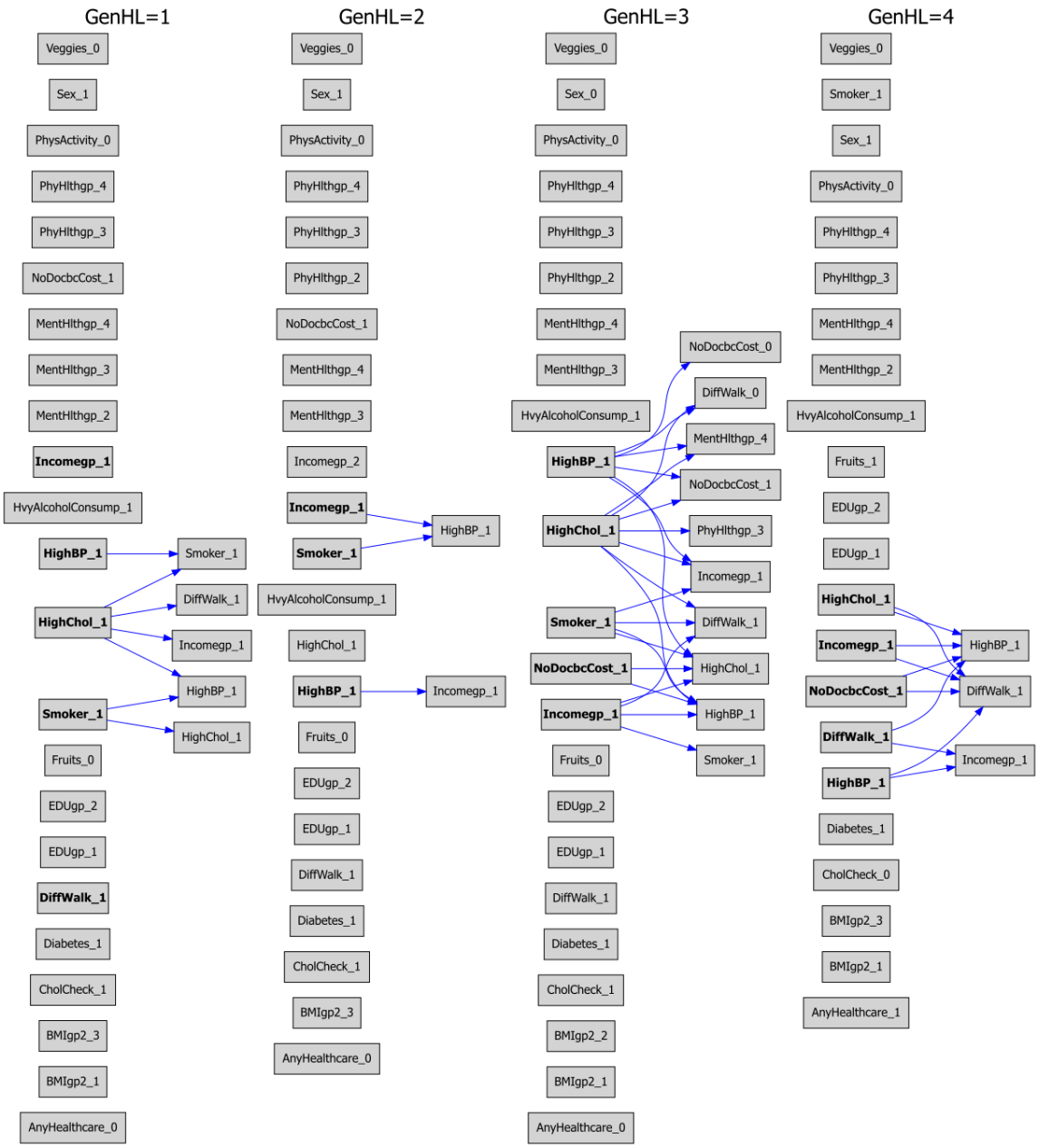

Figure 4: Age2's order-2 interacting effect evolution of positive chronic disease risk across GenHL: 1 to 4.

1.5 The Evolution of Age Group 4: Negative Disease Risk Patterns

The evolution of negative chronic disease risk in the Age = 4 subpopulation begins at the GenHL = 1 stratum, which bears similarity to the GenHL = 1 stratum of Age = 3, as illustrated in

Figure 5. However, the number of major 1-feature categories expands with the first appearance of $Fruits_1$, indicating subtle yet meaningful differentiation. We begin by listing the symmetric order-2 interactions (major 2-feature categories) observed at this stratum.

Symmetric Order-2 Interactions

1. $CholCheck_0$: couples with none.
2. Sex_0: couples with $PhyHlthgp_1$, $Diabetes_0$, $DiffWalk_0$, $Incompgp_3$, $HighBP_0$, $HighChol_0$.
3. $HighBP_0$: couples with $PhyHlthgp_1$, $Diabetes_0$, $DiffWalk_0$, Sex_0, $Smoker_0$, $HighChol_0$.
4. $HighChol_0$: couples with $Diabetes_0$, $DiffWalk_0$, $HighBP_0$, Sex_0, $Smoker_0$.
5. $Fruits_1$: couples with $EDUgp_3$, $Diabetes_0$, $DiffWalk_0$, $Incompgp_3$, $Smoker_0$.
6. $Incompgp_3$: couples with $Diabetes_0$, Sex_0, $Smoker_0$.
7. $Smoker_0$: couples with $Diabetes_0$, $DiffWalk_0$, $Incompgp_3$, $HighBP_0$, $HighChol_0$, $Fruits_1$, $BMIgp2_2$.
8. $BMIgp2_2$: couples with $DiffWalk_0$, $Smoker_0$.
9. $PhyHlthgp_1$: couples with Sex_0, $HighBP_0$.
10. $EDUgp_3$: couples with none.
11. $Diabetes_0$: couples with $DiffWalk_0$, $Incompgp_3$, $HighBP_0$, Sex_0, $Smoker_0$, $HighChol_0$, $Fruits_1$.
12. $DiffWalk_0$: couples with $Diabetes_0$, $HighBP_0$, Sex_0, $Smoker_0$, $HighChol_0$, $Fruits_1$, $BMIgp2_2$.

Directional Order-2 Interactions The following are directional order-2 interactions indicating features that provide additional information to enhance or modulate negative disease risk:

1. $CholCheck_0$: no incoming directional effect.
2. Sex_0: receives directional input from $PhysActivity_1$ only.
3. $HighBP_0$ and $HighChol_0$: no incoming directional effects.
4. $Fruits_1$: receives directional input from $PhysActivity_1$, $HighBP_0$, Sex_0, and $CholCheck_0$.
5. $Incompgp_3$: receives directional input from $BMIgp2_1$, $HighBP_0$, $HighChol_0$, and $Fruits_1$.
6. $Smoker_0$: enhanced by Sex_0 and $CholCheck_0$.
7. $BMIgp2_2$: receives input from $HighBP_0$, Sex_0, $HighChol_0$.
8. $PhyHlthgp_1$: directionally enhanced by $Diabetes_0$, $DiffWalk_0$, $HighBP_0$, Sex_0, $Smoker_0$, $HighChol_0$, $Fruits_1^*$ (double effect), $CholCheck_0$, and $BMIgp2_2$.
9. $EDUgp_3$: same input features as $PhyHlthgp_1$.
10. $Diabetes_0$: receives input from $CholCheck_0$ and $CholCheck_1$.
11. $DiffWalk_0$: receives input from $CholCheck_0$ and $CholCheck_1$.

Negative-to-Positive Risk Switching A distinct pattern at this stratum is the widespread switching from negative to positive disease risk, driven by several major 1-feature categories associated with positive risk. These switches are particularly relevant for understanding and preventing chronic disease:

1. $DiffWalk_1$ overturns: Sex_0, $HighBP_0$, $HighChol_0$, $Fruits_1$, $Smoker_0$, $EDUgp_3$, $PhyHlthgp_1$, and $Diabetes_0$.
2. $Diabetes_1$ overturns: $HighChol_0$, $Fruits_1$, $Incompgp_3$, $Smoker_0$, $PhyHlthgp_1$, $EDUgp_3$, and $DiffWalk_1$.
3. $HighBP_1$ overturns: $Fruits_1$, $Incompgp_3$, $Smoker_0$, $BMIgp2_2$, $PhyHlthgp_1$, $EDUgp_3$, $Diabetes_0$, and $DiffWalk_0$.
4. $HighChol_1$ overturns: $Fruits_1$, $Smoker_0$, $PhyHlthgp_1$, $EDUgp_3$, $Diabetes_0$, and $DiffWalk_0$.
5. $Smoker_1$ overturns: $PhyHlthgp_1$.
6. Sex_1 overturns: $Fruits_1$, $BMIgp2_2$, $PhyHlthgp_1$, $EDUgp_3$, $Diabetes_0$, and $DiffWalk_0$.
7. $Fruits_0$ overturns: $EDUgp_3$, $Diabetes_0$, and $DiffWalk_0$.

This expanded scope of negative-to-positive transitions is first observed at this GenHL = 1 stratum in the Age = 4 group and is particularly informative from a chronic disease prevention perspective.

For example, item [5] illustrates how the protective effects of physical activity ($PhyHlthgp_1$) can be completely negated by smoking ($Smoker_1$) at Age = 4. Similarly, item [6] underscores a sobering gender disparity: being male (Sex_1) appears to nullify multiple protective effects—such as those from eating fruits ($Fruits_1$), maintaining fitness ($BMIgp2_2$), engaging in physical activity ($PhyHlthgp_1$), higher education ($EDUgp_3$), diabetes control ($Diabetes_0$), and mobility ($DiffWalk_0$). On a more optimistic note, item [7] suggests that abstaining from fruits ($Fruits_0$) can still reverse several risk factors, illustrating the nuanced role of dietary habits.

GenHL = 2 The evolution of negative disease risk within the Age = 4 subpopulation enters the GenHL = 2 stratum, as visualized in the second column panel of Figure 5. Compared to the GenHL = 1 stratum, this panel appears slightly more complex in terms of the number of major 1-feature categories. Notably, this stratum marks the first appearance of $HvyAlcoholConsump_1$ as a major 1-feature category. The major 2-feature categories are summarized below, grouped into symmetric, directional, and switching effects.

Symmetric Order-2 Interactions

1. $CholCheck_0$ couples with $Diabetes_0$ only.
2. $HvyAlcoholConsump_1$ couples with $Diabetes_0$ only.
3. $HighBP_0$ couples with $HighChol_0$, $Diabetes_0$, Sex_0, $Smoker_0$.
4. $Incompgp_3$ couples with Sex_0 only.

5. $BMIgp2_2$ couples with $Smoker_0$, $DiffWalk_0$.
6. $Smoker_0$ couples with $Diabetes_0$, Sex_0, $HighBP_0$, $DiffWalk_0$, $BMIgp2_2$.
7. $DiffWalk_0$ couples with $Diabetes_0$, Sex_0, $Smoker_0$, $BMIgp2_2$.
8. $PhyHlthgp_1$ couples with $HighChol_0$, $Diabetes_0$, Sex_0.
9. $Diabetes_0$ couples with $HighChol_0$, Sex_0, $HighBP_0$, $Smoker_0$, $DiffWalk_0$, $HvyAlcoholConsump_1$, $PhyHlthgp_1$.
10. $HighChol_0$ couples with $HighBP_0$, $Diabetes_0$, $PhyHlthgp_1$.
11. $Veggies_1$ couples with none.
12. Sex_0 couples with $Diabetes_0$, $HighBP_0$, $Smoker_0$, $DiffWalk_0$, $PhyHlthgp_1$.

In general, the number and breadth of pairings (i.e., coupling lengths) are somewhat shorter than those observed in the GenHL = 1 stratum.

Directional Order-2 Interactions

1. On $CholCheck_0$: additional directional information is provided by $MentHlthgp_1$.
2. On $HvyAlcoholConsump_1$: by $HighChol_0$, Sex_0, and $HighBP_0$.
3. On $HighBP_0$: by $CholCheck_1$, $HighChol_1$, $CholCheck_0$, and $Smoker_1$.
4. On $Incompgp_3$: by $HighChol_0$, $Diabetes_0$, $CholCheck_1$, $CholCheck_0$, $HighBP_0$, and $Smoker_0$.
5. On $BMIgp2_2$: by $HighChol_0$, $Diabetes_0$, Sex_0, and $HighBP_0$.
6. On $Smoker_0$: by $CholCheck_1$, $CholCheck_0$, and $BMIgp2_3$.
7. On $DiffWalk_0$: by $HighChol_0$, $CholCheck_1$, $CholCheck_0$, $HighBP_0$, $HvyAlcoholConsump_0$, and $HvyAlcoholConsump_1$.
8. On $PhyHlthgp_1$: by $HighChol_0$, $Diabetes_0$, and Sex_0.
9. On $Diabetes_0$: by $CholCheck_1$ and $CholCheck_0$.
10. On $HighChol_0$: by $CholCheck_1$, Sex_0, $HighBP_1$, $Smoker_0$, $Smoker_1$, and $EDUgh_3$.
11. On $Veggies_1$: by $HighChol_0$, $Diabetes_0$, Sex_0, $CholCheck_0$, $HighBP_0$, $Smoker_0$, $DiffWalk_0$, $BMIgp2_2$, $HvyAlcoholConsump_1$, and $Incompgp_3$.
12. On Sex_0: by $BMIgp2_2$, $HvyAlcoholConsump_1$, and $Incompgp_3$.

Negative-to-Positive Switching Effects The following positive-risk features lead to negative-to-positive disease risk transitions. These switching effects closely resemble the patterns seen in GenHL = 1 but exhibit a more consistent and concentrated structure:

1. $HighChol_1$ overturns: $Incompgp_3$, $BMIgp2_2$, $DiffWalk_0$, $PhyHlthgp_1$, $Diabetes_0$, $Veggies_1$.
2. $Diabetes_1$ overturns: $Incompgp_3$, $BMIgp2_2$, $Smoker_0$, $DiffWalk_0$, $PhyHlthgp_1$, $Veggies_1$.
3. Sex_1 overturns: $Incompgp_3$, $BMIgp2_2$, $Smoker_0$, $DiffWalk_0$, $PhyHlthgp_1$, $Diabetes_0$, $Veggies_1$.
4. $HighBP_1$ overturns: $Incompgp_3$, $BMIgp2_2$, $Smoker_0$, $DiffWalk_0$, $PhyHlthgp_1$, $Diabetes_0$, $Veggies_1$.
5. $Smoker_1$ overturns: $DiffWalk_0$, $PhyHlthgp_1$, $Diabetes_0$, $Veggies_1$.
6. $DiffWalk_1$ overturns: $BMIgp2_2$, $PhyHlthgp_1$, $Diabetes_0$, $Veggies_1$, Sex_0.
7. $Incompgp_1$ overturns: $PhyHlthgp_1$, $Veggies_1$.

As seen in the first four items, key 1-feature categories of positive disease risk—$HighChol_1$, $Diabetes_1$, Sex_1, and $HighBP_1$—consistently overturn nearly the same seven 1-feature categories of negative disease risk. This concentrated and repeated switching pattern marks an important developmental milestone in the evolution of chronic disease risk. It is the first time such intensive and overlapping reversals are observed across the Age = 1 to Age = 3 subpopulations, underscoring the fragility of protective factors at this stage.

GenHL = 3 As expected, the negative disease risk evolution becomes more intricate and expanded upon entering the GenHL = 3 stratum of the Age = 4 subpopulation. As shown in the third column panel of Figure 5, this increase in complexity does not stem from a higher number of major 1-feature categories. Rather, it arises from an expanded network of major 2-feature interactions—both symmetric and directional—as well as an intensified pattern of negative-to-positive switching effects. These three categories of interaction are detailed below.

Symmetric Order-2 Interactions

1. $HvyAlcoholConsump_1$: couples with none.
2. $HighChol_0$: couples with $DiffWalk_0$, $Incompgp_3$, $Smoker_0$, Sex_0, $Diabetes_0$, $HighBP_0$.
3. $PhyHlthgp_1$: couples with $Smoker_0$, Sex_0.
4. $HighBP_0$: couples with $DiffWalk_0$, $Incompgp_3$, $Smoker_0$, Sex_0, $Diabetes_0$, $HighChol_0$.
5. $Veggies_1$: couples with none.
6. $Diabetes_0$: couples with $DiffWalk_0$, $Incompgp_3$, $Smoker_0$, Sex_0, $HighBP_0$, $HighChol_0$.
7. $BMIgp2_2$: couples with $DiffWalk_0$, $Incompgp_3$, $Smoker_0$, $Diabetes_0$.
8. $DiffWalk_0$: couples with $Smoker_0$, Sex_0, $Diabetes_0$, $HighBP_0$, $HighChol_0$, $BMIgp2_2$.

9. Sex_0: couples with $DiffWalk_0$, $Incompgp_3$, $Smoker_0$, $Diabetes_0$, $HighBP_0$, $HighChol_0$, $PhyHlthgp_1$.

10. $Incompgp_3$: couples with Sex_0, $Diabetes_0$, $HighBP_0$, $HighChol_0$, $BMIgp2_2$.

11. $Smoker_0$: couples with $DiffWalk_0$, Sex_0, $Diabetes_0$, $HighBP_0$, $HighChol_0$, $BMIgp2_2$, $PhyHlthgp_1$.

12. $CholCheck_0$: couples with none.

Directional Order-2 Interactions

1. On $HvyAlcoholConsump_1$: extra information from $HighBP_0$, $HighChol_0$, $CholCheck_0$.

2. On $HighChol_0$: from $Incompgp_1$, $Smoker_1$, $HighBP_1$, Sex_1, $CholCheck_0$, $CholCheck_1$.

3. On $PhyHlthgp_1$: from $CholCheck_0$, $DiffWalk_0$, $Diabetes_0$, $CholCheck_1$, $HighBP_0$, $HighChol_0$.

4. On $HighBP_0$: from $Smoker_1$, Sex_1, $HighChol_1$, $CholCheck_0$, $CholCheck_1$.

5. On $Veggies_1$: from $CholCheck_0$, $DiffWalk_0$, $Incompgp_3$, $Smoker_0$, Sex_0, $Diabetes_0$, $CholCheck_1$, $HighBP_0$, $HighChol_0$, $HvyAlcoholConsump_1$, $BMIgp2_2$, $PhyHlthgp_1$.

6. On $Diabetes_0$: from $CholCheck_0$, $CholCheck_1$, $BMIgp2_2$, $BMIgp2_3$.

7. On $BMIgp2_2$: from Sex_0, $HighBP_0$, $HighChol_0$.

8. On $DiffWalk_0$: from $CholCheck_0$, $CholCheck_1$, $AnyHealthcare_0$, $AnyHealthcare_1$, $HvyAlcoholConsump_1$, $HvyAlcoholConsump_0$, $BMIgp2_3$.

9. On Sex_0: from $HighChol_1$, $CholCheck_0$, $CholCheck_1$, $PhysActivity_0$, $PhysActivity_1$, $NoDocbcCost_0$.

10. On $Incompgp_3$: from $CholCheck_0$, $Smoker_0$, $CholCheck_1$, $HvyAlcoholConsump_1$, $HvyAlcoholConsump_0$.

11. On $Smoker_0$: from $CholCheck_1$, $BMIgp2_3$, $NoDocbcCost_0$.

12. On $CholCheck_0$: from $BMIgp2_3$ only.

Negative-to-Positive Switching Effects In the GenHL = 3 stratum of Age = 4, the pattern of negative-to-positive risk switching is more concentrated than in the previous stratum, indicating a form of contraction in the reversal landscape. All switching interactions target a smaller subset of negative-risk features, particularly $Veggies_1$ and $PhyHlthgp_1$.

1. $Incompgp_1$ overturns: $Veggies_1$.

2. $Diabetes_1$ overturns: $PhyHlthgp_1$, $Veggies_1$.

3. $Smoker_1$ overturns: $DiffWalk_0$, $PhyHlthgp_1$, $Veggies_1$.

4. $HighBP_1$ overturns: $PhyHlthgp_1$, $Veggies_1$, $Diabetes_0$, $BMIgp2_2$, $DiffWalk_0$.

5. Sex_1 overturns: $PhyHlthgp_1$, $Veggies_1$, $Diabetes_0$, $DiffWalk_0$.

6. $HighChol_1$ overturns: $PhyHlthgp_1$, $Veggies_1$, $Diabetes_0$, $DiffWalk_0$.

7. $DiffWalk_1$ overturns: $PhyHlthgp_1$, $Veggies_1$, $Diabetes_0$, $BMIgp2_2$, Sex_0.

This contraction in switching patterns marks a significant structural transition in the evolution of negative disease risk. While previous strata exhibited broader reversals, GenHL = 3 narrows its focus to a small cluster of vulnerable features, primarily $Veggies_1$, $PhyHlthgp_1$, and $DiffWalk_0$, which are consistently overturned by a core set of dominant positive-risk categories. This signals increasing instability and fragility in protective factors as general health worsens within this age group.

GenHL = 4 A further and significant contraction is observed as the evolution of negative disease risk enters the GenHL = 4 stratum of the Age = 4 subpopulation, as shown in the fourth column panel of Figure 5. Not only does the number of major 1-feature categories shrink, but the collection of major 2-feature categories also contracts considerably.

Symmetric Order-2 Interactions

1. $AnyHealthcare_0$ couples with $Incompgp_3$ only.

2. Sex_0 couples with $Incompgp_3$, $HighBP_0$, $HighChol_0$, $Smoker_0$, $Diabetes_0$, and $DiffWalk_0$.

3. $DiffWalk_0$ couples with $HighBP_0$, $HighChol_0$, $Smoker_0$, and $DiffWalk_1$.

4. $Incompgp_3$ couples with Sex_0 only.

5. $CholCheck_0$ couples with $HighBP_0$ and $Diabetes_0$.

6. $Smoker_0$ couples with $HighBP_0$, $HighChol_0$, $Diabetes_0$, and $DiffWalk_0$.

7. $Diabetes_0$ couples with $HighBP_0$, $HighChol_0$, $Smoker_0$, Sex_0, and $CholCheck_0$.

8. $HighChol_0$ couples with $HighBP_0$, $HighChol_0$, $Smoker_0$, $Diabetes_0$, and $DiffWalk_0$.

9. $HighBP_0$ couples with $HighChol_0$, $Smoker_0$, $Diabetes_0$, Sex_0, $DiffWalk_0$, and $CholCheck_0$.

This contraction has the effect of eliminating many secondary 1-feature categories previously active in earlier strata, such as $Veggies_1$, $Fruits_1$, and others. The contraction pattern is similarly reflected in the directional interactions.

Directional Order-2 Interactions

1. On $AnyHealthcare_0$: additional information is provided by $HighBP_0$ only.

2. On Sex_0: by $AnyHealthcare_1$ and $NoDocbcCost_0$.

3. On $DiffWalk_0$: by $AnyHealthcare_1$, $AnyHealthcare_0$, and Sex_0.

4. On $Incompgp_3$: by $AnyHealthcare_0$, $HighBP_0$, $HighChol_0$, $Smoker_0$, and Sex_0.

5. On $CholCheck_0$: no directional effects.

6. On $Smoker_0$: by Sex_0, $BMIgp2_3$, and $HighBP_1$.

7. On $Diabetes_0$: by $Incompgp_3$, $NoDocbcCost_0$, $DiffWalk_0$, and $CholCheck_0$.

8. On $HighChol_0$: by Sex_0, $BMIgp2_3$, $HighBP_1$, $BMIgp2_2$, Sex_1, $Smoker_1$, and $DiffWalk_1$.

9. On $HighBP_0$: by $CholCheck_1$, Sex_1, $HighChol_1$, $Smoker_1$, and $DiffWalk_1$.

It is particularly striking that while both $HighBP_0$ and $HighChol_0$ receive directional inputs from well-known positive disease risk categories, including $HighBP_1$, $HighChol_1$, Sex_1, $Smoker_1$, and $DiffWalk_1$, their negative disease risk is not overturned. This critical phenomenon, observed in both GenHL = 3 and GenHL = 4, underscores an important implication: at this stage, certain protective factors become resilient against interference from dominant positive-risk traits. In contrast, $Diabetes_0$ remains vulnerable, with its protective effect overturned by Sex_1 and $HighChol_1$, the only two instances of negative-to-positive switching in this stratum.

GenHL = 5 Upon reaching the GenHL = 5 stratum in the Age = 4 subpopulation—shown in the fifth column panel of Figure 5, the evolution of negative disease risk reaches its most contracted state. No further negative-to-positive switchings are observed, and the overall interaction structure becomes increasingly sparse.

Symmetric Order-2 Interactions

1. $HighBP_0$ couples with $Diabetes_0$ and $HighChol_0$.

2. $HighChol_0$ couples with $DiffWalk_0$, $Diabetes_0$, $HighBP_0$, $Incompgp_3$.

3. $Smoker_0$ couples with $DiffWalk_0$ (two arrows), $Diabetes_0$, $HighBP_0$, $Incompgp_3$, $HighChol_0$.

4. $BMIgp2_2$: couples with none.

5. Sex_0 couples with $DiffWalk_0$, $Diabetes_0$, and $Incompgp_3$.

6. $Incompgp_3$ couples with $DiffWalk_0$, $Diabetes_0$, $HighChol_0$, $Smoker_0$, and Sex_0.

7. $Diabetes_0$ (extra arrow): couples with $HighBP_0$, $Incompgp_3$, $HighChol_0$, $Smoker_0$, and Sex_0.

8. $DiffWalk_0$ couples with $Incompgp_3$, $HighChol_0$, $Smoker_0$, and Sex_0.

Directional Order-2 Interactions

1. On $HighBP_0$: by $HighChol_1$, $Incompgp_1$.

2. On $HighChol_0$: by $Incompgp_1$, $DiffWalk_1$, and $HighBP_1$.

3. On $Smoker_0$: no additional directional effects.

4. On $BMIgp2_2$: by $DiffWalk_0$, $Diabetes_0$, $HighBP_0$, $Incompgp_3$, $HighChol_0$, and $Smoker_0$.

5. On Sex_0: by $HighBP_0$, $HighChol_0$, $Smoker_0$.

6. On $Incompgp_3$: no directional inputs.

7. On $Diabetes_0^*$: by $EDUgp_3$ only.

8. On $DiffWalk_0$: by $HighBP_0$ only.

This final stage of contraction in the Age = 4 subpopulation can be interpreted as the loss of chronic disease resistance. From GenHL = 3 to GenHL = 5, the number of interactive negative-risk effects steadily declines, indicating a diminishing capacity to leverage combinations of protective traits for preventive synergy.

Conversely, the earlier expansion observed from GenHL = 1 to GenHL = 3 highlights the broad and diverse ways in which the human condition can become imbalanced. These imbalances—reflected in secondary or lifestyle-related risks—underscore a critical asymmetry in chronic disease dynamics: the human system is more readily destabilized than stabilized. This insight, revealed by the structural evolution across GenHL strata, is a foundational observation in understanding chronic disease susceptibility and prevention.

1.6 The Evolution of Age 4: Positive

GenHL=1 We now examine the evolution of positive disease risk within the Age=4 subpopulation along the GenHL-axis. As shown in the first column-panel of Figure 6, the initial stratum (GenHL=1) displays a high degree of complexity. The symmetric order-2 interacting effects are itemized below:

1. $BMIgp2_3$ couples with $Diabetes_1$ only.

2. $Incompgp_1$ couples with $EDUgp_2$, $HighBP_1$ and $HighChol_1$.

3. $Smoker_1$ couples with $EDUgp_2$, Sex_1 and $HighBP_1$.

4. $EDUgp_2$ couples with $Smoker_1$ only.

5. $Fruits_0$ couples with $HighBP_1$ only.

6. $CholCheck_1$ couples with Sex_1 only.

7. $Diabetes_1$ couples with $DiffWalk_1$ only.

8. $HighChol_1$ couples with Sex_1 and $HighBP_1$.

9. $HighBP_1$ couples with Sex_1, $HighChol_1$, $Fruits_0$, $Incompgp_1$ and $Smoker_1$.

10. Sex_1 couples with $HighBP_1$, $HighChol_1$ and $Smoker_1$.

11. $DiffWalk_1$ couples with $Diabetes_1$.

The coupling lengths in these symmetric order-2 effects are relatively short, indicating that few major 1-feature-categories play equal roles. In contrast, directional order-2 effects are far more extensive, as listed below:

1. $BMIgp2_3$, directional extra information is provided by Sex_1, $HighBP_1$, $HighChol_1$, $DiffWalk_1$, $Fruits_0$ and $Incompgp_1$.

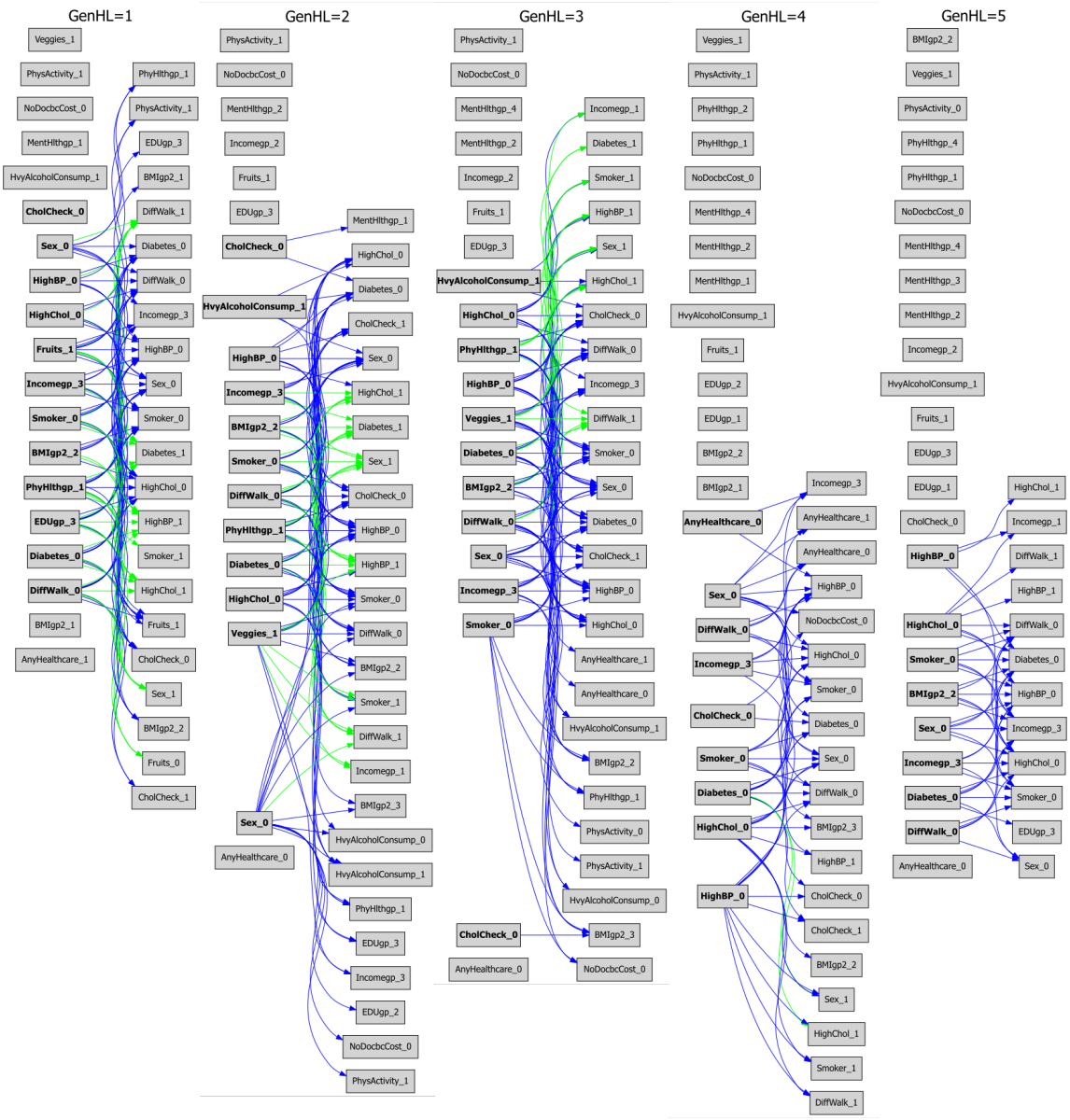

Figure 5: Age4's order-2 interacting effect evolution of negative chronic disease risk across GenHL: 1 to 5.

2. $Incompgp_1$, directional extra information is provided by Sex_1, $HighBP_1$, $HighChol_1$, $DiffWalk_1$ and $Diabetes_1$.

3. $Smoker_1$, directional extra information is provided by $HighChol_1$, $DiffWalk_1$, $Diabetes_1$ and $Incompgp_1$.

4. $EDUgp_2$, directional extra information is provided by Sex_1, $HighBP_1$, $HighChol_1$, $DiffWalk_1$ and $Diabetes_1$.

5. $Fruits_0$, directional extra information is provided by Sex_1, $DiffWalk_1$, $Diabetes_1$ and $DiffWalk_0$.

6. $CholCheck_1$, directional extra information is provided by $EDUgp_2$, $HighBP_1$, $HighChol_1$, $DiffWalk_1$, $Diabetes_1$, $PhysActivity_0$, $BMIgp2_3$, $Fruits_0$, $Incompgp_1$ and $Smoker_1$.

7. $Diabetes_1$, directional extra information is provided by $BMIgp2_3$ and $Smoker_1$.

8. $HighChol_1$, directional extra information is provided by $DiffWalk_1$, $Diabetes_1$, $Fruits_0$, $Incompgp_1$, $DiffWalk_0$, $EDUgp_3$, $Fruits_1$ and $Diabetes_0$.

9. $HighBP_1$, directional extra information is provided by $DiffWalk_1$, $Diabetes_1$, $DiffWalk_0$, $Smoker_0$, $Fruits_1$, $MentHlthgp_3$ and $Diabetes_0$.

10. Sex_1, directional extra information is provided by $DiffWalk_1$, $Diabetes_1$, $Incompgp_1$, $DiffWalk_0$, $Fruits_1$, $Diabetes_0$ and $CholCheck_1$.

11. $DiffWalk_1$, directional extra information is provided by $Diabetes_1$ and $Diabetes_0$.

These extended directional interactions suggest that each major 1-feature-category of positive disease risk contributes uniquely, sometimes enhancing and sometimes reducing, depending on the combination of features and subpopulation involved.

Notably, only three relatively minor 1-feature-categories exhibit positive-to-negative switching:

1 $BMIgp2_3$ is overturned by $HighChol_0$, Sex_0, and $HighBP_0$.

2 $CholCheck_1$ is overturned by Sex_0, $HighBP_0$, $Incompgp_3$, $BMIgp2_3$, $DiffWalk_0$, $Smoker_0$, $Fruits_1$, and $Diabetes_0$.

3 Sex_1 is overturned by $CholCheck_0$.

GenHL=2 Upon entering the GenHL=2 stratum of Age=4, a significant expansion of positive disease risk interaction is observed, as demonstrated in the second column-panel of Figure 6. The number of major 2-feature-categories increases substantially, and their coupling patterns become notably more intricate.

Two isolated 1-feature-categories, $HvyAlcoholConsump_0$ and $EDUgp_1$, do not participate in these interactions and are therefore excluded. The remaining coupling structure is as follows:

1 Sex_1 couples with $CholCheck_1$, $Diabetes_1$, $DiffWalk_1$, $Incompgp_1$, $Smoker_1$, $HighChol_1$ and $HighBP_1$.

2 $Smoker_1$ couples with $CholCheck_1$, $Diabetes_1$, $DiffWalk_1$, $Incompgp_1$,$HighChol_1$, $HighBP_1$, Sex_1 and $Veggies_0$.

3 $HighBP_1$ couples with $Diabetes_1$, $DiffWalk_1$, $PhyHlthgp_3$, $Incompgp_1$, $Smoker_1$, $HighChol_1$ and Sex_1.

4 $Incompgp_1$ couples with $CholCheck_1$, $Diabetes_1$, $DiffWalk_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$, and Sex_1.

5 $BMIgp2_3$ couples with $CholCheck_1$ only.

6 $HighChol_1$ couples with $CholCheck_1$, $Diabetes_1$, $DiffWalk_1$, $PhyHlthgp_3$, $Incompgp_1$, $Smoker_1$, $HighBP_1$, Sex_1 and $Veggies_0$.

7 $Veggies_0$ couples with $PhyHlthgp_3$, $Smoker_1$ and $HighChol_1$.

8 $DiffWalk_1$ couples with $Diabetes_1$, $Incompgp_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$ and Sex_1.

9 $CholCheck_1$ couples with $Incompgp_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$, Sex_1 and $BMIgp2_3$.

10 $Diabetes_1$ couples with $DiffWalk_1$, $Incompgp_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$ and Sex_1.

11 $PhyHlthgp_3$ couples with $HighChol_1$, $HighBP_1$ and $Veggies_0$.

These interactions suggest that dominant 1-feature-categories in this stratum mutually reinforce one another, likely leading to substantial cumulative positive disease risks.

Directional order-2 interacting effects are likewise abundant:

1 upon Sex_1, directional extra information is provided by $DiffWalk_0$, $PhysHlthgp_1$, $Diabetes_0$, $Incompgp_3$ and $Smoker_0$.

2 upon $Smoker_1$, directional extra information is provided by $PhysHlthgp_4$, $Veggies_1$, $DiffWalk_0$, $PhysHlthgp_1$, $Diabetes_0$, $HvyAlcoholConsump_0$ and $PhyHlthgp_3$.

3 upon $HighBP_1$, directional extra information is provided by $DiffWalk_0$, $PhysHlthgp_1$, $Diabetes_0$, $HvyAlcoholConsump_0$, $Incompgp_3$ and $Smoker_0$.

4 upon $Incompgp_1$, directional extra information is provided by $PhysHlthgp_1$ only

5 upon $BMIgp2_3$, directional extra information is provided by $Diabetes_1$, $DiffWalk_1$, $PhyHlthgp_3$, $Incompgp_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$, Sex_1 and $EDUgp_1$.

6 upon $HighChol_1$, directional extra information is provided by $EDUgp_1$ only.

7 upon $Veggies_0$, directional extra information is provided by $Diabetes_1$, $DiffWalk_1$, dominant major 1-feature-categories and Sex_1 .

8 upon $DiffWalk_1$, directional extra information is provided by $Diabetes_0$ and Sex_0.

9 $CholCheck_1$, directional extra information is provided by $HvyAlcoholConsump_0$, $Diabetes_1$, $DiffWalk_1$, $PhyHlthgp_3$, $EDUgp_1$ and $Veggies_0$.

10 upon $Diabetes_1$, directional extra information is provided by $Smoker_1$ only.

11 $PhyHlthgp_3$, directional extra information is provided by Sex_1 only.

A notable pattern emerges: dominant positive disease risk categories generally receive additional directional information from secondary ones, but rarely the reverse. This asymmetry reflects the hierarchical structure of feature influence.

In this stratum, for the first time, several dominant 1-feature-categories of positive disease risk—such as $HighBP_1$ and $HighChol_1$—are overturned by their negative-risk counterparts ($HighBP_0$, $HighChol_0$), as summarized below:

1 $DiffWalk_0$ overturns $CholCheck_1$.

2 $CholCheck_0$ overturns Sex_1, $Smoker_1$ and $BMIgp2_3$.

3 $Diabetes_0$ overturns $BMIgp2_3$ and $CholCheck_1$.

4 $Incompgp_3$ overturns $CholCheck_1$ only.

5 $HighChol_0$ overturns $Smoker_1$, $HighBP_1$, $BMIgp2_3$, $Veggies_0$ and $CholCheck_1$.

6 $Smoker_0$ overturns $BMIgp2_3$ and $CholCheck_1$.

7 $HighBP_0$ overturns $Smoker_1$, $BMIgp2_3$, $HighChol_1$, $Veggies_0$, and $CholCheck_1$.

8 Sex_0 overturns $Smoker_1$, $BMIgp2_3$, $HighChol_1$ and $CholCheck_1$.

9 $BMIgp2_2$ overturns $CholCheck_1$ only.

10 $HvyAlcoholConsump_1$ overturns $CholCheck_1$ only.

Particularly surprising is the consistent reversal of $HighChol_1$ by Sex_0, which recurs in the next stratum (GenHL=3). This contrast is striking because Sex_1 otherwise acts as a strong enhancer in positive risk when coupled with other dominant categories like $HighBP_1$ and $HighChol_1$.

GenHL=3 The feature category of Sex continues to emerge as a critical factor in chronic disease dynamics along the Age-axis. This importance is further amplified in the GenHL=3 stratum of Age=4, as demonstrated in the third column-panel of Figure 6. Indeed, the evolution of positive disease risk patterns in GenHL=3 shows considerable overlap with those in GenHL=2, as revealed through the following itemized lists.

Symmetric Order-2 Interactions:

1. $CholCheck_1$ couples with $BMIgp2_3$, $HvyAlcoholConsump_0$, $Incompgp_1$, $Smoker_1$, $HighChol_1$ and Sex_1.

2. $HvyAlcoholConsump_0$ couples with $CholCheck_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$ and Sex_1.

3. $BMIgp2_3$ couples with $CholCheck_1$ only.

4. $Veggies_0$ couples with none.

5. Sex_1 couples with $CholCheck_1$, $Diabetes_1$, $DiffWalk_1$, $Incompgp_1$, $Smoker_1$, $HighChol_1$ and $HighBP_1$.

6. $Smoker_1$ couples with $HvyAlcoholConsump_0$, $CholCheck_1$, $Diabetes_1$, $DiffWalk_1$, $Incompgp_1$,$HighChol_1$, $HighBP_1$ and Sex_1.

7. $HighBP_1$ couples with $HvyAlcoholConsump_0$, $Diabetes_1$, $DiffWalk_1$, $Incompgp_1$, $Smoker_1$, $HighChol_1$ and Sex_1.

8. $HighChol_1$ couples with $HvyAlcoholConsump_0$, $CholCheck_1$, $Diabetes_1$, $DiffWalk_1$, $Incompgp_1$, $Smoker_1$, $HighBP_1$ and Sex_1.

9. $Incompgp_1$ couples with $CholCheck_1$, $Diabetes_1$, $DiffWalk_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$, and Sex_1.

10. $DiffWalk_1$ couples with $Diabetes_1$, $Incompgp_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$ and Sex_1.

11. $Diabetes_1$ couples with $DiffWalk_1$, $Incompgp_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$ and Sex_1.

12. $PhyHlthgp_4$ couples with none.

These symmetric couplings demonstrate a densely connected network among dominant 1-feature-categories, reinforcing the mutual enhancement of positive disease risk.

Directional Order-2 Interactions:

1. upon $CholCheck_1$, directional extra information is provided by $PhyHlthgp_4$, $Diabetes_1$ and $DiffWalk_1$.

2. $HvyAlcoholConsump_0$, directional extra information is provided by $BMIgp2_3$, $PhyHlthgp_4$, $Diabetes_1$, $DiffWalk_1$ and $Incompgp_1$.

3. $BMIgp2_3$, directional extra information is provided by$PhyHlthgp_4$, $CholCheck_1$, $Diabetes_1$, $DiffWalk_1$, $Incompgp_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$ and Sex_1.

4. $Veggies_0$, directional extra information is provided by $CholCheck_0$, $Diabetes_1$, $DiffWalk_1$, $Incompgp_1$, $Smoker_1$, $HighChol_1$, $HighBP_1$ and Sex_1.

5. Sex_1, directional extra information is provided by $Diabetes_0$, $DiffWalk_0$, $HvyAlcoholConsump_0$ and $PhyHlthgp_1$.

6. $Smoker_1$, directional extra information is provided by $DiffWalk_0$ only

7. $HighBP_1$, directional extra information is provided by $DiffWalk_0$, $Diabetes_0$ and $PhyHlthgp_1$.

8. $HighChol_1$, directional extra information is provided by $DiffWalk_0$, $Diabetes_0$, $PhyHlthgp_1$, $NoDocbcCost_0$ and $NoDocbcCost_1$.

9. $Incompgp_1$, directional extra information is provided by none.

10. $DiffWalk_1$, directional extra information is provided by Sex_0 and $Diabetes_0$.

11. $Diabetes_1$, directional extra information is provided by $PhyHlthgp_1$ only.

12. $PhyHlthgp_4$, directional extra information is provided by $HighChol_1$, $HighBP_1$, Sex_1 and $PhysActivity_0$.

This structure again reveals that dominant features such as Sex_1, $HighBP_1$, and $HighChol_1$ continue to be central hubs, receiving and transmitting extensive directional information from both primary and secondary features. A few variables, like $Incompgp_1$, provide no additional directional information, indicating a more passive role.

Positive-to-Negative Switchings:

1. $HvyAlcoholConsump_1$ overturns $CholCheck_1$.

2. $BMIgp2_2$ overturns $CholCheck_1$ and $HvyAlcoholConsump_0$.

3. $Incompgp_3$ overturns $CholCheck_1$, $HvyAlcoholConsump_0$ and $BMIgp2_3$.

4. $Smoker_0$ overturns $CholCheck_1$, $BMIgp2_3$ and $Veggies_0$.
item[5.] Sex_0 overturns $CholCheck_1$, $HvyAlcoholConsump_0$, $BMIgp2_3$, $Veggies_0$ and $HighChol_1$.

6. $CholCheck_0$ overturns $HvyAlcoholConsump_0$, $BMIgp2_3$, $Veggies_0$, Sex_1 and $Smoker_1$.

7. $DiffWalk_0$ overturns $CholCheck_1$, $HvyAlcoholConsump_0$ and $BMIgp2_3$.

8. $HighChol_0$ overturns $CholCheck_1$, $HvyAlcoholConsump_0$, $BMIgp2_3$, $Veggies_0$, Sex_1 and $Smoker_1$, $HighBP_1$ and $Incompgp_1$.

9. $HighBP_0$ overturns $CholCheck_1$, $HvyAlcoholConsump_0$, $BMIgp2_3$, $Veggies_0$, Sex_1 and $Smoker_1$, $HighChol_1$ and $Incompgp_1$.

10. $Diabetes_0$ overturns $CholCheck_1$, $HvyAlcoholConsump_0$ and $BMIgp2_3$.

11. $PhyHlthgp_1$ overturns $CholCheck_1$ and $HvyAlcoholConsump_0$.

These switchings signal the significant capacity of certain negative-risk 1-feature-categories to neutralize or reverse the effects of dominant positive-risk features. Particularly noteworthy are the reversals of $HighChol_1$ and $HighBP_1$ by their negative-risk counterparts ($HighChol_0$, $HighBP_0$), and the consistent disruptive role of Sex_0 and $DiffWalk_0$.

GenHL=4 As we transition into the GenHL=4 stratum of Age=4, the fourth column-panel of Figure 6 reveals a noticeable contraction in both the number and density of major 2-feature-category interactions. Despite this simplification, key structural patterns from the previous strata remain intact. For example, dominant positive-risk features such as $HighChol_1$ and $HighBP_1$ are still overturned by their negative-risk counterparts $HighBP_0$ and $HighChol_0$, respectively—highlighting a continued dynamic of competitive reversal.

The detailed symmetric and directional order-2 interactions for GenHL=4, along with updated positive-to-negative switchings, follow in the subsequent itemized list.

1. $AnyHealthcare_1$ couples with $Incompgp_1$, $HighChol_1$ and $Smoker_1$.

2. Sex_1 couples with $Incompgp_1$, $Diabetes_1$, $HighBP_1$, $DiffWalk_1$, $HighChol_1$ and $Smoker_1$.

3. $MentHlthgp_3$ couples with $Incompgp_1$, $Diabetes_1$ and $HighBP_1$.

4. $Smoker_1$ couples with $Incompgp_1$, $Diabetes_1$, $HighBP_1$, $AnyHealthcare_1$, $DiffWalk_1$, $HighChol_1$ and Sex_1.

5. $HighChol_1$ couples with $Incompgp_1$, $Diabetes_1$, $HighBP_1$, $AnyHealthcare_1$, $DiffWalk_1$, $Smoker_1$ and Sex_1.

6. $DiffWalk_1$ couples with $Diabetes_1$, $HighBP_1$, $HighChol_1$, $Smoker_1$ and Sex_1.

7. $Incompgp_1$ couples with $HighChol_1$, $Smoker_1$, Sex_1 and $MentHlthgp_3$.

8. $HighBP_1$ couples with $Diabetes_1$, $DiffWalk_1$, $HighChol_1$, $Smoker_1$, Sex_1 and $MentHlthgp_3$.

9. $Diabetes_1$ couples with $HighBP_1$, $DiffWalk_1$, $HighChol_1$, $Smoker_1$, Sex_1 and $MentHlthgp_3$.

1. upon $AnyHealthcare_1$, directional extra information is provided by $CholCheck_1$, $BMIgp2_3$, $Diabetes_1$, $HighBP_1$, $DiffWalk_1$, Sex_1 and $MentHlthgp_3$.
2. upon Sex_1, directional extra information is provided by $Diabetes_0$ and $MentHlthgp_1$.
3. upon $MentHlthgp_3$, directional extra information is provided by $DiffWalk_1$, $HighChol_1$ and $Smoker_1$.
4. upon $Smoker_1$, directional extra information is provided by none.
5. upon $HighChol_1$, directional extra information is provided by none.
6. upon $DiffWalk_1$, directional extra information is provided by none.
7. upon $Incompgp_1$, directional extra information is provided by $Diabetes_1$, $HighBP_1$ and $DiffWalk_1$.
8. upon $HighBP_1$, directional extra information is provided by none.
9. upon $Diabetes_1$, directional extra information is provided by $PhyHlthgp_3$.

1. All members of the following set of 1-feature-categories of negative risks: $\{Incompgp_3, Sex_0, CholCheck_0, Diabetes_0$ individually overturn $AnyHealthcare_1$.
2. $DiffWalk_0$ overturns $Incompgp_1$.
3. $HighBP_0$ overturns $AnyHealthcare_1$, Sex_1, $MentHlthgp_3$, $Smoker_1$, $HighChol_1$, $DiffWalk_1$ and $Incompgp_1$.
4. $HighChol_0$ overturns $AnyHealthcare_1$, Sex_1, $Smoker_1$, $DiffWalk_1$, $Incompgp_1$ and $HighBP_1$.
5. $Smoker_0$ overturns $AnyHealthcare_1$, $Incompgp_1$ and $HighBP_1$.

GenHL=5 As shown in the 5th column-panel of Figure 6, a further and substantial contraction is observed. The overall density of both blue and green color-coded arrows is markedly sparse, indicating a significant reduction in the complexity of positive disease risk interactions. Notably, Sex_1 no longer appears as a major 1-feature-category within this stratum. The few remaining major 1-feature-categories associated with positive disease risk—six in total—are all well-established and widely recognized in the chronic disease literature.

1. $HighBP_1$ couples with $Smoker_1$, $HighChol_1$ and $DiffWalk_1$.
2. $Diabetes_1$ couples with $DiffWalk_1$ only.
3. $Incompgp_1$ couples with $Smoker_1$ and $HighChol_1$.
4. $DiffWalk_1$ couples with $HighChol_1$, $Diabetes_1$ and $HighBP_1$.
5. $Smoker_1$ couples with $HighBP_1$ and $Incompgp_1$.
6. $HighChol_1$ couples with $DiffWalk_1$, $HighBP_1$ and $Incompgp_1$.

1. upon $HighBP_1$, directional extra information is provided by none.

2. upon $Diabetes_1$, directional extra information is provided by $HighChol_1$.

3. upon $Incompgp_1$, directional extra information is provided by $HighBP_1$.

4. upon $DiffWalk_1$, directional extra information is provided by none.

5. upon $Smoker_1$, directional extra information is provided by $HighChol_1$ and $DiffWalk_1$.

6. upon $HighChol_1$, directional extra information is provided by none.

This GenHL=5 stratum indeed sheds an important light on chronic disease dynamics: with the exception of $Diabetes_1$, each of the remaining five major 1-feature-categories of positive disease risk can be overturned by corresponding major 1-feature-categories associated with negative disease risk.

1. $HighChol_0$ overturns $HighBP_1$, $Incompgp_1$, $DiffWalk_1$ and $Smoker_1$.

2. $Diabetes_0$ overturns $DiffWalk_1$.

3. $HighBP_0$ overturns $Incompgp_1$, $DiffWalk_1$, $Smoker_1$ and $HighChol_1$.

Similar to the Age=3 subpopulation, four heatmaps displaying the individual risk landscapes across all computed major 2-feature-categories from GenHL=1 to GenHL=4 for Age=4 are presented in Figure 7. The waxing and waning patterns of negative and positive disease risks become even more pronounced across the four panels along the GenHL-axis. To better understand the stark contrasts between negative and positive disease risks, beyond the effects of medical treatment and variations in chronic disease latency among older individuals, we must consider the potential confounding factors embedded in the definition of GenHL.

In general, human perception of health is closely tied to functional ability in daily life. Individuals who remain active and capable of managing most everyday tasks may be inclined to self-report as having good general health, even in the presence of clinical indicators such as high blood pressure, elevated cholesterol, or diabetes, so long as they have not experienced overt symptoms like stroke or heart disease. Evidence supporting this confounding mechanism emerges from two contrasting patterns observed in the heatmaps.

In panel (A), a substantial number of atypical subjects with positive disease risks appear in GenHL=1, and this number steadily declines along the GenHL-axis. In contrast, panel (D) reveals a growing number of atypical subjects with negative disease risks as we move toward GenHL=4. This progressive increase from GenHL=1 to GenHL=4 highlights how subjective assessments of general health may obscure the presence of underlying medical risks, particularly within the older Age=4 subpopulation.

1.7 The Evolution of Age 5: Negative

GenHL=1 We now present the evolution of negative disease risk within the Age=5 subpopulation along the GenHL-axis, beginning with the GenHL=1 stratum as shown in the first column-panel of Figure 8. Compared to the GenHL=1 stratum of Age=4, this stratum exhibits a noticeably lower level of complexity. Moreover, several unusual patterns emerge. For instance, $HighBP_0$ retains only one arrow and is no longer coupled with $HighChol_0$, marking a distinct decoupling from prior interactions observed in younger age strata. In fact, many dominant 1-feature-categories

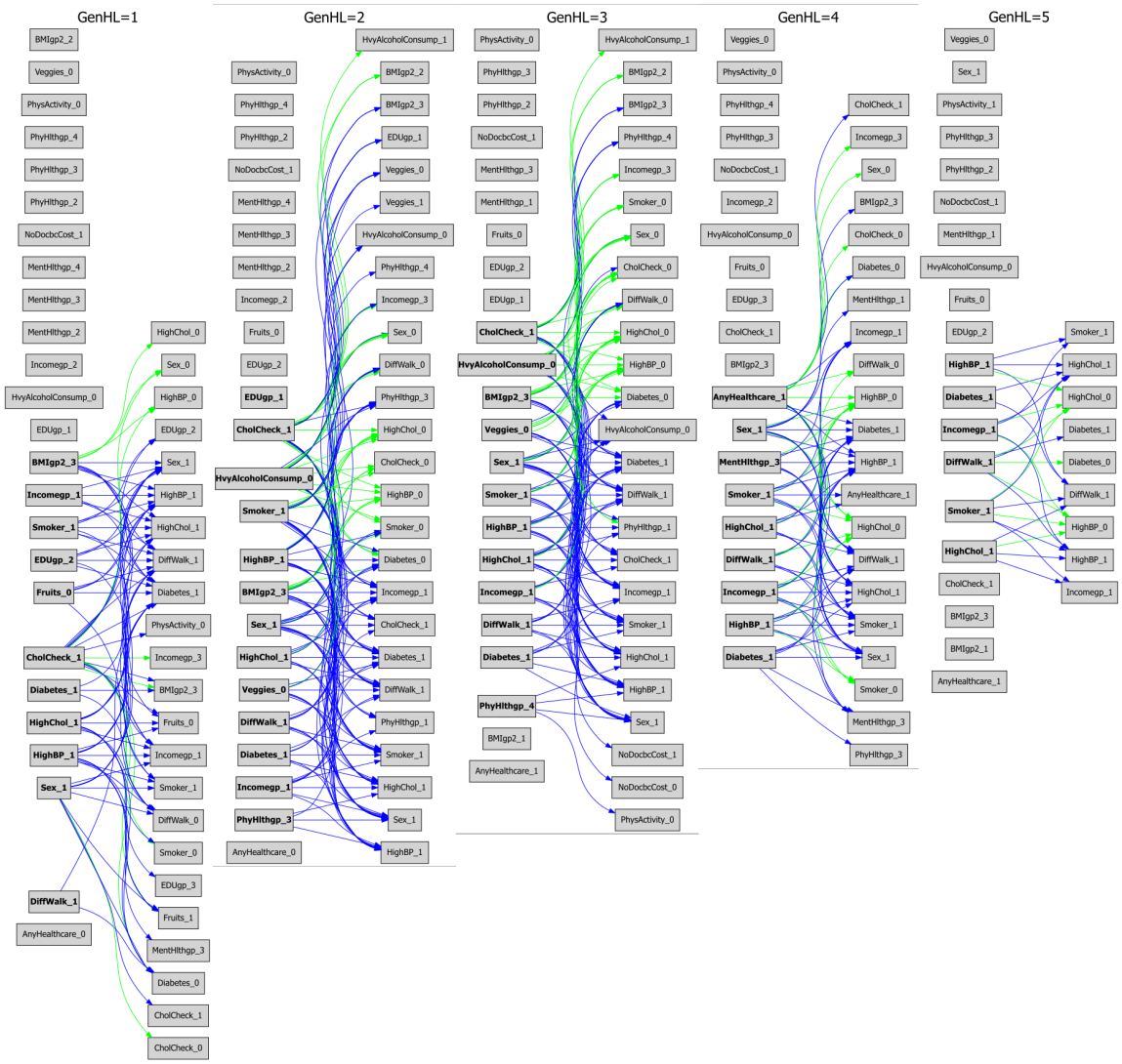

Figure 6: Age4's order-2 interacting effect evolution of positive chronic disease risk across GenHL: 1 to 5.

that played key roles in the Age=3 and Age=4 subpopulations no longer participate in symmetric order-2 interacting effects.

Below, we itemize the symmetric order-2 interacting effects identified within this stratum:

1. $HighBP_0$ couples with Sex_0 only.
2. $BMIgp2_2$ couples with Sex_0 and $DiffWalk_0$.
3. $HighChol_0$ couples with Sex_0, $DiffWalk_0$ and $PhyHlthgp_1$.

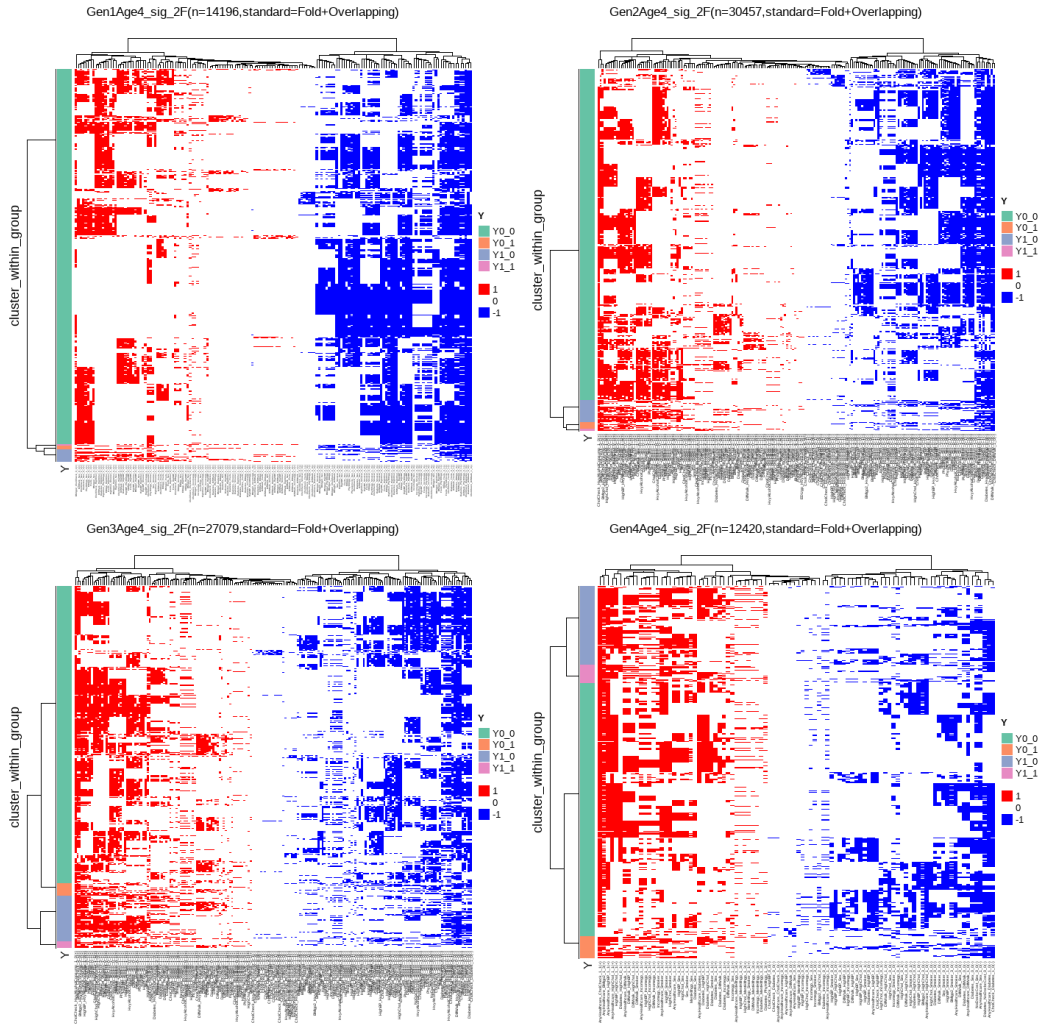

Figure 7: Heatmaps of individual risk-landscape: GenHL=1(A), GenHL=2(B), GenHL=3(C) and GenHL=4(D) of Age4.

4. $Diabetes_0$ couples with $Veggies_1$, Sex_0 and $MentHlthgp_1$.
5. $Smoker_0$ couples with $Veggies_1$, $MentHlthgp_1$, $PhyHlthgp_1$ and $PhysActivity_1$.
6. $PhyHlthgp_1$ couples with $Veggies_1$, Sex_0, $MentHlthgp_1$, $HighChol_0$ and $PhysActivity_1$.
7. $Veggies_1$ couples with $PhyHlthgp_1$, $Smoker_0$ and $Diabetes_0$.
8. $PhysActivity_1$ couples with Sex_0, $MentHlthgp_1$, $PhyHlthgp_1$ and $Smoker_0$.
9. $DiffWalk_0$ couples with Sex_0, $MentHlthgp_1$, $HighChol_0$ and $BMIgp2_2$.

10. $MentHlthgp_1$ coupled with $DiffWalk_0$, $PhyHlthgp_1$, $Smoker_0$, $PhysActivity_1$ and $Diabetes_0$.
11. Sex_0 couples with $DiffWalk_0$, $PhyHlthgp_1$, $HighChol_0$, $PhysActivity_1$, $Diabetes_0$ and $BMIgp2_2$.

In the itemized list above, Sex_0 clearly emerges as the most influential 1-feature-category of negative disease risk. It forms order-2 interacting effects with a broader collection of major 1-feature-categories than do $HighBP_0$, $HighChol_0$, or any others in this stratum. Notably, within the Age=5 subpopulation, the features $Veggies_1$, $PhyHlthgp_1$, $PhysActivity_1$, and $MentHlthgp_1$ become prominent for the first time in the observed evolution of negative disease risk, indicating a possible age-related shift in the landscape of chronic disease resistance. The directional order-2 interacting effects are itemized below.

1. upon $HighBP_0$, directional extra information is provided by none.
2. $BMIgp2_2$, directional extra information is provided by $HighBP_0$ only.
3. $HighChol_0$, directional extra information is provided by $HighBP_0$ only.
4. $Diabetes_0$, directional extra information is provided by $DiffWalk_0$, $HighBP_0$ and $HighChol_0$.
5. $Smoker_0$, directional extra information is provided by Sex_0, $DiffWalk_0$, $HighBP_0$, and $HighChol_0$.
6. $PhyHlthgp_1$, directional extra information is provided by $DiffWalk_0$, $HighBP_0$ and $Smoker_0$.
7. $Veggies_1$, directional extra information is provided bySex_0, $DiffWalk_0$, $DiffWalk_1$, $HighBP_0$, $MentHlthgp_1$, $HighBP_1$, Sex_1, and $HighChol_0$.
8. $PhysActivity_1$, directional extra information is provided by $DiffWalk_0$, $HighBP_0$, $HighChol_0$, and $BMIgp2_2$.
9. $DiffWalk_0$, directional extra information is provided by $HighBP_0$ only.
10. $MentHlthgp_1$, directional extra information is provided by Sex_0, $HighBP_0$, $HighChol_0$, and $BMIgp2_2$.
11. Sex_0, directional extra information is provided by $HighBP_0$ and $Incomegp_3$.

The directional enhancer role of $HighBP_0$ is wide, but unusual.

There are three major 1-feature-categories of positive disease risks, including Sex_1, that could cause many negative-to-positive switchings.

1. $DiffWalk_1$ overturns $BMIgp2_2$, $HighChol_0$, $Diabetes_0$, $Smoker_0$, $PhyHlthgp_1$, $PhysActivity_1$, and Sex_0.
2. $HighBP_1$ overturns $Diabetes_0$, $Smoker_0$, $PhyHlthgp_1$, $PhysActivity_1$, $DiffWalk_0$, and $MentHlthgp_1$.
3. Sex_1 overturns $Diabetes_0$, $PhyHlthgp_1$, $PhysActivity_1$, $DiffWalk_0$, and $MentHlthgp_1$.
4. $MentHlthgp_3$ overturns $PhysActivity_1$.
5. $Diabetes_1$ overturns $MentHlthgp_1$.

The message of Sex_1 overturning 5 major 1-feature-categories is worth further thinking and investigating.

GenHL=2 Upon entering the GenHL=2 stratum, the evolution of negative disease risk contracts and exhibits slightly reduced complexity compared to the preceding GenHL=1 stratum, as shown in the second column-panel of Figure 8. Notably, $HighBP_0$ appears to regain its usual dominant status as a major 1-feature-category. In addition, $PhyHlthgp_1$ begins to exhibit characteristics of a dominant negative disease risk factor, with $PhysActivity_1$ also emerging as influential to a certain extent.

1. $PhyHlthgp_1$ couples with $Smoker_0$, $HighChol_0$, $BMIgp2_2$, $HighBP_0$, $Diabetes_0$, and $DiffWalk_0$.
2. $DiffWalk_0$ couples with $Smoker_0$, $HighChol_0$, $BMIgp2_2$, $HighBP_0$, Sex_0, $Diabetes_0$, and $PhyHlthgp_1$.
3. $PhysActivity_1$ couples with Sex_0 only.
4. $Diabetes_0$ couples with $Smoker_0$, $HighChol_0$, $BMIgp2_2$, $HighBP_0$, Sex_0, $DiffWalk_0$, and $PhyHlthgp_1$.
5. $Smoker_0$ couples with $BMIgp2_2$, $Diabetes_0$, $DiffWalk_0$, and $PhyHlthgp_1$.
6. $BMIgp2_2$ couples with $Smoker_0$, $HighChol_0$, $HighBP_0$, $Diabetes_0$, $DiffWalk_0$, and $PhyHlthgp_1$.
7. $HighChol_0$ couples with $BMIgp2_2$, $HighBP_0$, $Diabetes_0$, $DiffWalk_0$, and $PhyHlthgp_1$.
8. $HighBP_0$ couples with $HighChol_0$, $BMIgp2_2$, Sex_0, $Diabetes_0$, $DiffWalk_0$, and $PhyHlthgp_1$.
9. Sex_0 couples with $HighBP_0$, $Diabetes_0$, $DiffWalk_0$, and $PhysActivity_1$.

It is worth noting that Sex_0 continues to serve as an important major 1-feature-category, consistently coupling with key dominant negative disease risk factors to form order-2 interacting effects.

1. upon $PhyHlthgp_1$, directional extra information is provided by none.
2. upon $DiffWalk_0$, directional extra information is provided by none.
3. upon $PhysActivity_1$, directional extra information is provided by $Smoker_0$, $HighChol_0$, $HighBP_0$, $Diabetes_0$, $DiffWalk_0$, and $PhyHlthgp_1$.
4. upon $Diabetes_0$, directional extra information is provided by none.
5. $Smoker_0$, directional extra information is provided by $HighChol_0$, $HighBP_0$ and Sex_0.
6. $BMIgp2_2$, directional extra information is provided by Sex_0 only.
7. $HighChol_0$, directional extra information is provided by $HighBP_1$, Sex_0 and $BMIgp2_3$.
8. $HighBP_0$, directional extra information is provided by $BMIgp2_3$.
9. Sex_0, directional extra information is provided by $PhyHlthgp_1$ and $MentHlthgp_1$.

Due to the high prevalence of symmetric order-2 interacting effects, the number of directional interactions is relatively small and limited in scope.

1. $HighChol_1$ overturns $PhyHlthgp_1$, $DiffWalk_0$, $PhysActivity_1$, and $Diabetes_0$.
2. $Diabetes_1$ overturns $PhyHlthgp_1$, $DiffWalk_0$, $PhysActivity_1$, and $Smoker_0$.
3. $HighBP_1$ overturns $PhyHlthgp_1$, $DiffWalk_0$, $PhysActivity_1$, and $Diabetes_0$.
4. Sex_1 overturns $PhyHlthgp_1$, $DiffWalk_0$, $PhysActivity_1$, $Diabetes_0$, $Smoker_0$, and $BMIgp2_2$.
5. $DiffWalk_1$ overturns $PhyHlthgp_1$, $PhysActivity_1$, $Diabetes_0$, $Smoker_0$, and $BMIgp2_2$.

These five key 1-feature-categories of positive disease risk overturns dominant 1-feature-categories of negative disease risks.

GenHL=3 Further contraction is observed in the 3rd column-panel of Figure 8. It is important to reiterate the central role of Sex_0, which appears to exert even greater influence than other typically dominant 1-feature-categories.

1. $DiffWalk_0$ couples with Sex_0, $HighChol_0$, $Smoker_0$, and $Diabetes_0$.
2. $PhyHlthgp_1$ couples with $Smoker_0$ and $Diabetes_0$.
3. $Diabetes_0$ couples with Sex_0, $HighChol_0$, $HighBP_0$, $Smoker_0$, $DiffWalk_0$, and $PhyHlthgp_1$.
4. $HighBP_0$ couples with Sex_0, $HighChol_0$, $Smoker_0$, and $Diabetes_0$.
5. $HighChol_0$ couples with Sex_0, $HighBP_0$, $Smoker_0$, $Diabetes_0$, and $DiffWalk_0$.
6. Sex_0 couples with $HighChol_0$, $HighBP_0$, $Smoker_0$, $Diabetes_0$, $DiffWalk_0$, and $PhyHlthgp_1$.
7. $Cholcheck_0$ couples with none.

Further contraction is evident in the itemized list of directional major 2-feature-categories below. Only a limited number of 1-feature-categories are capable of providing additional information.

1. $DiffWalk_0$, directional extra information is provided by $Cholcheck_0$, $Cholcheck_1$, and $HighBP_0$.
2. $PhyHlthgp_1$, directional extra information is provided by $Cholcheck_0$, $Cholcheck_1$, Sex_0, $HighChol_0$, $HighBP_0$, and $DiffWalk_0$.
3. $Diabetes_0$, directional extra information is provided by $Cholcheck_0$ and $Cholcheck_1$.
4. $HighBP_0$, directional extra information is provided by none.
5. $HighChol_0$, directional extra information is provided by $HighBP_1$.
6. Sex_0, directional extra information is provided by $HighBP_1$ and $Incomegp_3$.
7. $Cholcheck_0$, directional extra information is provided by none.

It is noteworthy that within this GenHL stratum of the most advanced age group, $HighBP_1$ is unable to overturn the effects of $HighChol_0$ and Sex_0.

The list of negative-to-positive switchings has markedly diminished compared to the previous stratum.

1. $HighChol_1$ overturns $DiffWalk_0$, $PhyHlthgp_1$, and $Diabetes_0$.
2. Sex_1 overturns $DiffWalk_0$, $PhyHlthgp_1$, and $Diabetes_0$.
3. $Somker_1$ overturns $DiffWalk_0$, $PhyHlthgp_1$, and $Diabetes_0$.

The three key 1-feature-categories of positive disease risks each overturn the same set of three major 1-feature-categories of negative disease risks.

GenHL=4 A drastic contraction is observed in the 4th column-panel of Figure 8. It is important to reiterate the significance of the five remaining major 1-feature-categories—most notably, Sex_0. Among them, Sex_0 appears to play an even more central role than the other four: $Smoker_0$, $DiffWalk_0$, $HighChol_0$, and $HighBP_0$.

1. Sex_0 couples with $DiffWalk_0$, $HighChol_0$ and $HighBP_0$.
2. $Smoker_0$ couples with none.
3. $DiffWalk_0$ couples with Sex_0 only.
4. $HighChol_0$ couples with Sex_0 only.
5. $HighBP_0$ couples with Sex_0 only.

1. Sex_0, directional extra information is provided by none.
2. $Smoker_0$, directional extra information is provided by couples with $HighChol_0$, $HighBP_0$, and Sex_0.
3. $DiffWalk_0$, directional extra information is provided by $HighChol_0$.
4. $HighChol_0$, directional extra information is provided by $HighBP_0$.
5. $HighBP_0$, directional extra information is provided by none.

The directional order-2 interacting effects become notably sparse in the GenHL=4 stratum, and no switchings are observed. Upon entering the GenHL=5 stratum, the evolution of negative disease risk nearly vanishes, with only two remaining major 1-feature-categories—$HighChol_0$ and $HighBP_0$—and no observable order-2 interactions.

From the overall perspective of Figure 8, the evolution of negative disease risk dynamics clearly peaks at the GenHL=1 and GenHL=2 strata, followed by a sharp contraction at GenHL=3, near disappearance at GenHL=4, and complete cessation at GenHL=5. This trajectory of contraction raises important implications. Within the context of the disease outcome categories $\mathcal{Y} = (HD, STK)$, the initial diversity of major 1-feature-categories of negative disease risks—along with the richness of their interactions—emerges as a collective and dominant force in shaping chronic disease dynamics. The eventual collapse of this structure in the most advanced GenHL stratum signals a critical weakening of the population's intrinsic resistance mechanisms.

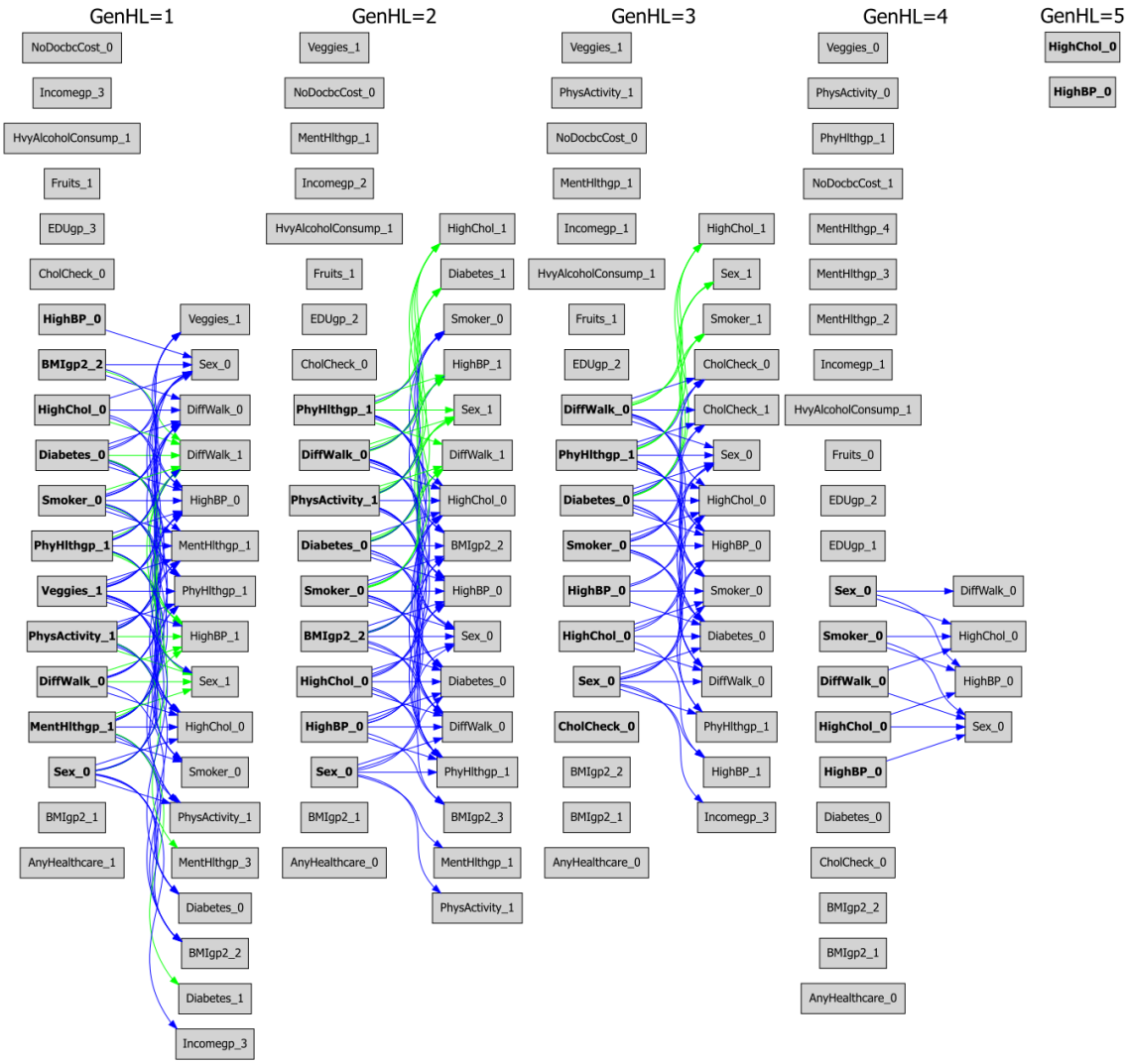

Figure 8: Age5's order-2 interacting effect evolution of negative chronic disease risk across GenHL: 1 to 5.

1.8 The Evolution of Age 5: Positive

GenHL=1 We present the evolution of positive disease risk within the Age=5 subpopulation along the GenHL-axis. The GenHL=1 stratum, as shown in the first column-panel of Figure 9, demonstrates a considerable degree of complexity. Below, we itemize the symmetric order-2 interacting effects:

1. $BMIgp2_3$ couples with $HighChol_1$ only.
2. $Smoker_1$ couples with none.

3. $HighChol_1$ couples with $BMIgp2_3$ only.
4. $PhyHlthgp_3$ couples with $HighBP_1$ only.
5. $DiffWalk_1$ couples with $HighBP_1$ only.
6. $Diabetes_1$ couples with none.
7. $PhysActivity_0$ couples with $HighBP_1$ only.
8. Sex_1 couples with $HighBP_1$ only
9. $HighBP_1$ couples with Sex_1, $DiffWalk_1$, $PhyHlthgp_3$, and $PhysActivity_0$.

This list turns out much simpler than that of GenHL=1 of Age=4.

1. upon $BMIgp2_3$, directional extra information is provided by $Diabetes_1$, $Veggies_0$, $HighBP_1$ and $DiffWalk_1$.
2. upon $Smoker_1$, directional extra information is provided by $Diabetes_1$, $Veggies_0$, Sex_1, $HighBP_1$, and $DiffWalk_1$.
3. upon $HighChol_1$, directional extra information is provided by $Diabetes_1$, Sex_1, $HighBP_1$, $DiffWalk_1$, and $PhyHlthgp_3$.
4. upon $PhyHlthgp_3$, directional extra information is provided by none.
5. upon $DiffWalk_1$, directional extra information is provided by none.
6. upon $Diabetes_1$, directional extra information is provided by $HighBP_1$ and $DiffWalk_1$.
7. upon $PhysActivity_0$, directional extra information is provided by $DiffWalk_1$ only.
8. upon Sex_1, directional extra information is provided by $DiffWalk_1$, $PhyHlthgp_1$, $DiffWalk_0$, $MentHlthgp_1$, and $MentHlthgp_3$.
9. upon $HighBP_1$, directional extra information is provided by $PhyHlthgp_1$, $DiffWalk_0$, $MentHlthgp_1$, $MentHlthgp_3$, $PhysActivity_0$, and $PhysActivity_1$.

The dominant major 1-feature-categories of positive disease risk in this stratum are less coherent compared to those observed in the previous age subpopulations. Notably, $HighBP_1$ behaves distinctly from $HighChol_1$ and Sex_1: it forms numerous symmetric order-2 interacting effects with other major 1-feature-categories while also receiving additional directional information from several non-major 1-feature-categories. In contrast, $HighChol_1$ and Sex_1 form fewer symmetric interactions but allow both major and non-major 1-feature-categories to contribute substantial directional information.

Only three switchings are observed in the first column-panel of Figure 9.

1. $HighBP_0$ overturns $BMIgp2_3$ and $Smoker_1$.
2. $HvyAlcoholConsump_1$ overturns $Veggies_0$.

This limited number of switchings appears to be a defining characteristic of the GenHL=1 stratum within the Age=5 subpopulation.

GenHL=2 As the evolution of positive disease risk progresses into the GenHL=2 stratum, it remains equally sparse as observed in GenHL=1.

1. $Smoker_1$ couples with $HighBP_1$ only.
2. $BMIgp2_3$ couples with none.
3. $PhyHlthgp_2$ couples with Sex_1 only.
4. $DiffWalk_1$ couples with $HighChol_1$, $HighBP_1$ and Sex_1.
5. $PhysActivity_0$ couples with none.
6. $HighBP_1$ couples with $HighChol_1$, Sex_1, $Smoker_1$ and $DiffWalk_1$.
7. $HighChol_1$ couples with $HighBP_1$, Sex_1 and $DiffWalk_1$.
8. Sex_1 couples with $HighChol_1$, $HighBP_1$, $PhyHlthgp_2$ and $DiffWalk_1$.
9. $Diabetes_1$ couples with none.

1. upon $Smoker_1$, directional extra information is provided by $Diabetes_1$, $HighChol_1$, and Sex_1.
2. upon $BMIgp2_3$, directional extra information is provided by $Diabetes_1$, $HighChol_1$, $HighBP_1$, Sex_1, $PhyHlthgp_2$, and $DiffWalk_1$.
3. upon $PhyHlthgp_2$, directional extra information is provided by $Diabetes_1$, $HighChol_1$, and $HighBP_1$.
4. upon $DiffWalk_1$, directional extra information is provided by $Diabetes_0$ only.
5. upon $PhysActivity_0$, directional extra information is provided by $HighChol_1$, $HighBP_1$, Sex_1, and $DiffWalk_1$.
6. upon $HighBP_1$, directional extra information is provided by $Diabetes_1$, $Diabetes_0$, $PhyHlthgp_1$, and $DiffWalk_0$.
7. upon $HighChol_1$, directional extra information is provided by $Diabetes_1$, $Diabetes_0$, $PhyHlthgp_1$, and $DiffWalk_0$.
8. upon Sex_1, directional extra information is provided by $PhyHlthgp_1$, and $DiffWalk_0$.
9. upon $Diabetes_1$, directional extra information is provided by Sex_1, $DiffWalk_1$, $DiffWalk_0$, and $Incomegp_1$.

The contrasting phenomenon of very few symmetric interactions but relatively numerous directional ones reveals a distinctive mechanism underlying chronic disease dynamics in the most advanced age subpopulation. This pattern suggests a converging spectrum of both causal and preventive factors in chronic disease, where the complexity of risk accumulation is countered by a similarly multifaceted set of preventive influences.

The number of positive-to-negative switchings is five, significantly fewer than the number of negative-to-positive switchings observed within the same GenHL=2 stratum. This asymmetry reflects a key feature embedded in the aforementioned special mechanism underlying chronic disease dynamics in the most advanced age subpopulation.

1. Sex_0 overturns $BMIgp2_3$ and $Smoker_1$.
2. $HighChol_0$ overturns $HighBP_1$, $BMIgp2_3$, and $Smoker_1$.
3. $HighBP_0$ overturns $BMIgp2_3$ only.

GenHL=3 The evolution of positive disease risk contracts upon reaching the GenHL=3 stratum of Age=5. This contraction is evident in the overall reduction of order-2 interacting effects, or major 2-feature-categories. However, in contrast to this general trend, the number of symmetric order-2 interacting effects expands significantly, as seen in the itemized list below. This pattern—of contraction in breadth but expansion in symmetric structure—is a distinctive characteristic of GenHL=3 across all age subpopulations.

1. $PhyHlthgp_3$ couples with none.
2. $DiffWalk_1$ couples with $HighChol_1$, $HighBP_1$, and Sex_1.
3. $Smoker_1$ couples with $HighChol_1$, $HighBP_1$, $Diabetes_1$, and Sex_1.
4. Sex_1 couples with $HighChol_1$, $HighBP_1$, $Diabetes_1$, $DiffWalk_1$, and $Smoker_1$.
5. $Cholcheck_1$ couples with none.
6. $Diabetes_1$ couples with Sex_1, $DiffWalk_1$, and $Smoker_1$.
7. $HighBP_1$ couples with $HighChol_1$, Sex_1, $DiffWalk_1$, and $Smoker_1$.
8. $HighChol_1$ couples with $HighBP_1$, Sex_1, $DiffWalk_1$, and $Smoker_1$.

1. upon $PhyHlthgp_3$, directional extra information is provided by $HighChol_1$ only.
2. upon $DiffWalk_1$, directional extra information is provided by $Diabetes_1$ only.
3. upon $Smoker_1$, directional extra information is provided by $Diabetes_0$, $DiffWalk_0$, and $DiffWalk_1$.
4. upon Sex_1, directional extra information is provided by $DiffWalk_0$ only.
5. upon $Cholcheck_1$, directional extra information is provided by $HighChol_1$, $HighBP_1$, $Diabetes_1$, $PhyHlthgp_3$, Sex_1, $DiffWalk_1$, and $Smoker_1$.
6. upon $Diabetes_1$, directional extra information is provided by $HighChol_1$ only.
7. upon $HighBP_1$, directional extra information is provided by $Diabetes_1^*$ (2 arrows).
8. upon $HighChol_1$, directional extra information is provided by $DiffWalk_0$, $NoDocbcCost_0$ and $NoDocbcCost_1$.

1. $Cholcheck_1$ is overturned by $Diabetes_0$, $DiffWalk_0$, $HighBP_0$, $Smoker_0$, Sex_0, $HighChol_0$ and $PhyHlthgp_1$.
2. $HighBP_1$ is overturned by Sex_0 and $HighChol_0$.

GenHL=4 The evolution of positive disease risk contracts significantly upon reaching the GenHL=4 stratum of Age=5. This contraction mirrors that observed in the evolution of negative disease risk within the same stratum.

1. $DiffWalk_1$ couples with $HighChol_1$, $HighBP_1$, and Sex_1.
2. $Smoker_1$ couples with none.
3. $HighChol_1$ couples with Sex_1 and $DiffWalk_1$.
4. $HighBP_1$ couples with Sex_1 and $DiffWalk_1$.
5. Sex_1 couples with $HighChol_1$ and $DiffWalk_1$.

1. $DiffWalk_1$, directional extra information is provided by none.
2. $Smoker_1$, directional extra information is provided by $HighBP_1$, Sex_1, $HighChol_1$ and $DiffWalk_1$.
3. $HighChol_1$, directional extra information is provided by none.
4. $HighBP_1$, directional extra information is provided by $HighChol_1$ only.
5. Sex_1, directional extra information is provided by none.

No switchings are observed in the GenHL=4 stratum. The evolution of positive disease risk effectively ends at GenHL=5, with only one remaining major 1-feature-category, $HighBP_1$, and no major 2-feature-categories. This conveys a clear message about the converging nature of positive disease risk: within the GenHL=5 stratum of the Age=5 subpopulation, $HighBP_1$ stands as the sole confirmed positive risk. In contrast, the converging status of negative disease risk within the same stratum identifies $HighBP_0$ and $HighChol_0$ as the only confirmed negative risks.

Together, these findings reflect a critical reality of the Age=5 population: the ratio of diseased to non-diseased individuals no longer exhibits the imbalance characteristic of younger age groups (e.g., Age=1 and Age=2). Instead, all individuals in this advanced age subpopulation can be viewed as survivors, resilient to all potential risk features except for a core set—namely, $HighBP$, $HighChol$, Sex, $Smoker$, and $DiffWalk$.

From Figure 10, we continue to observe the mutually interacting wax-and-wane patterns of negative and positive disease risk evolution across the series of heatmaps depicting individual risk landscapes against the computed major 2-feature-categories from GenHL=1 to GenHL=3 within the Age=5 subpopulation. The contrasts between the two types of atypical subjects—those exhibiting high positive vs. high negative disease risk—become increasingly pronounced. This observation brings to mind the popular adage, “Keep active and stay busy," often cited as a lifestyle guideline for individuals in advanced age.

However, the persistent fact that non-diseased individuals remain the majority within each stratum from GenHL=1 to GenHL=3 points to a critical insight of this study: The current response variable $\mathcal{Y} = (STK, HD)$ may not be sufficient to fully capture and represent the broader landscape of chronic disease dynamics in U.S. society. This suggests that the scope of $\mathcal{Y}$ should potentially be expanded to include other chronic conditions such as diabetes, cancer, and additional long-term health challenges that play significant roles in the chronic disease profile of this population.

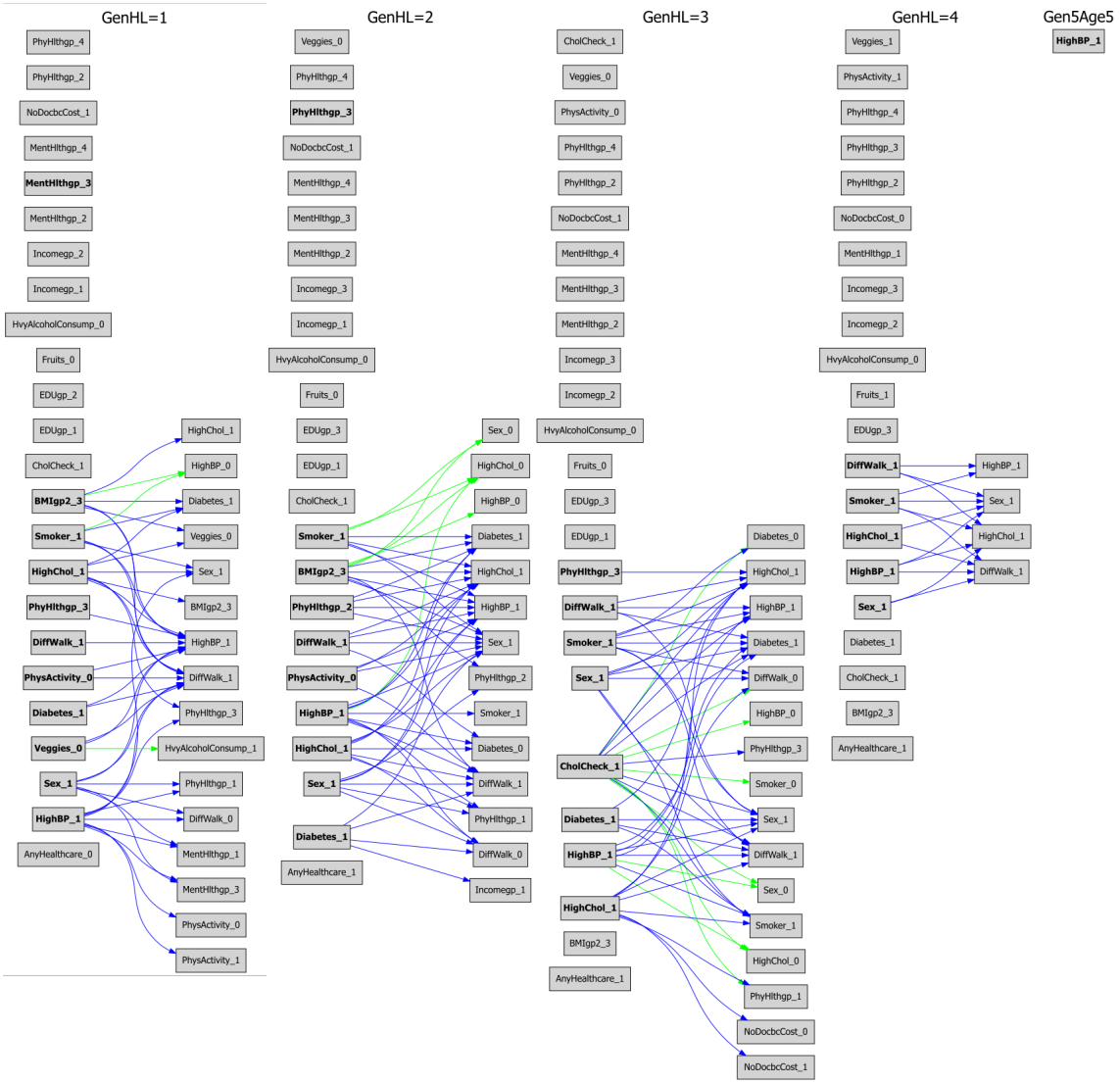

Figure 9: Age5's order-2 interacting effect evolution of positive chronic disease risk across GenHL: 1 to 5.

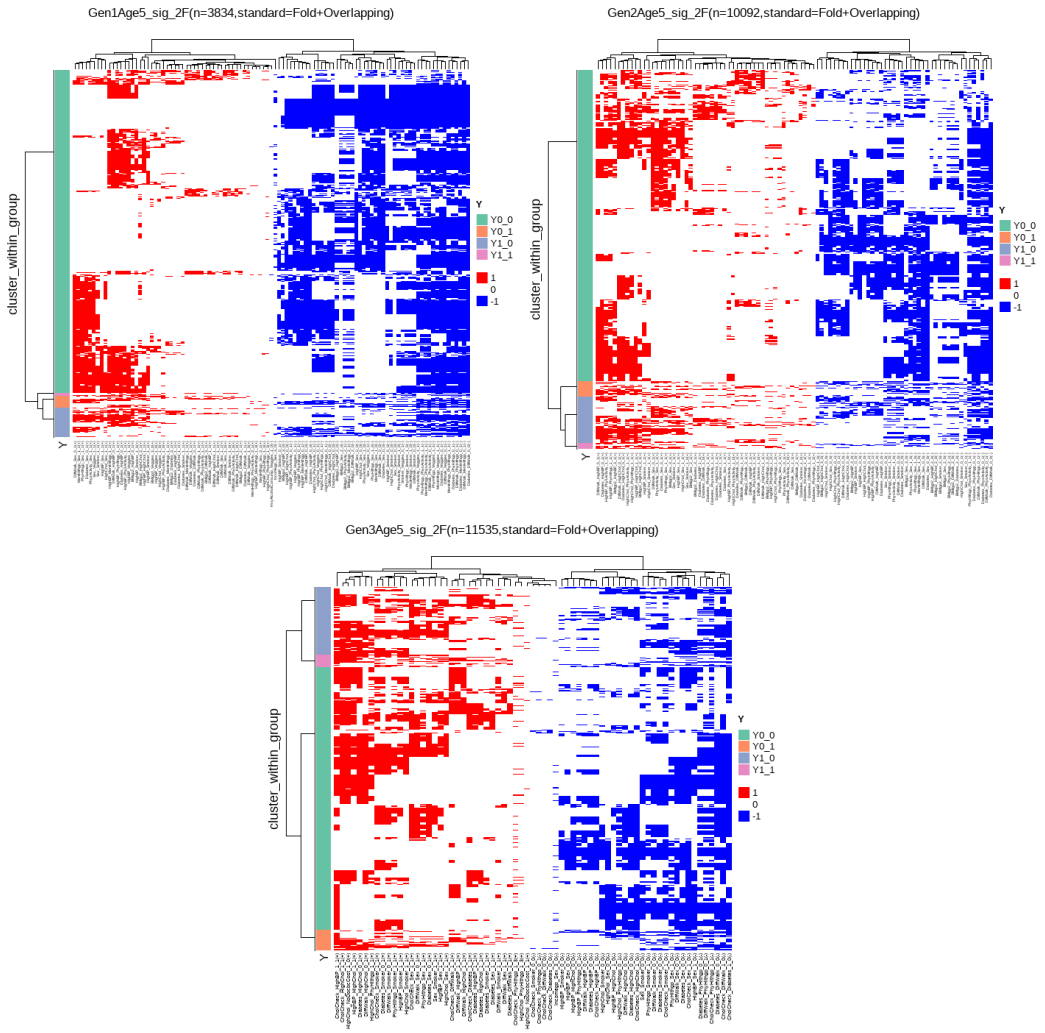

Figure 10: Heatmaps of individual risk-landscape: GenHL=1(A), GenHL=2(B) and GenHL=3(C) of Age5..